

Video Info

File name: P-CDL-I-M6-1-Scaling with Google Cloud Operations

Presenter:



06

Scaling with Google Cloud Operations

Module 6

Scaling with Google Cloud Operations

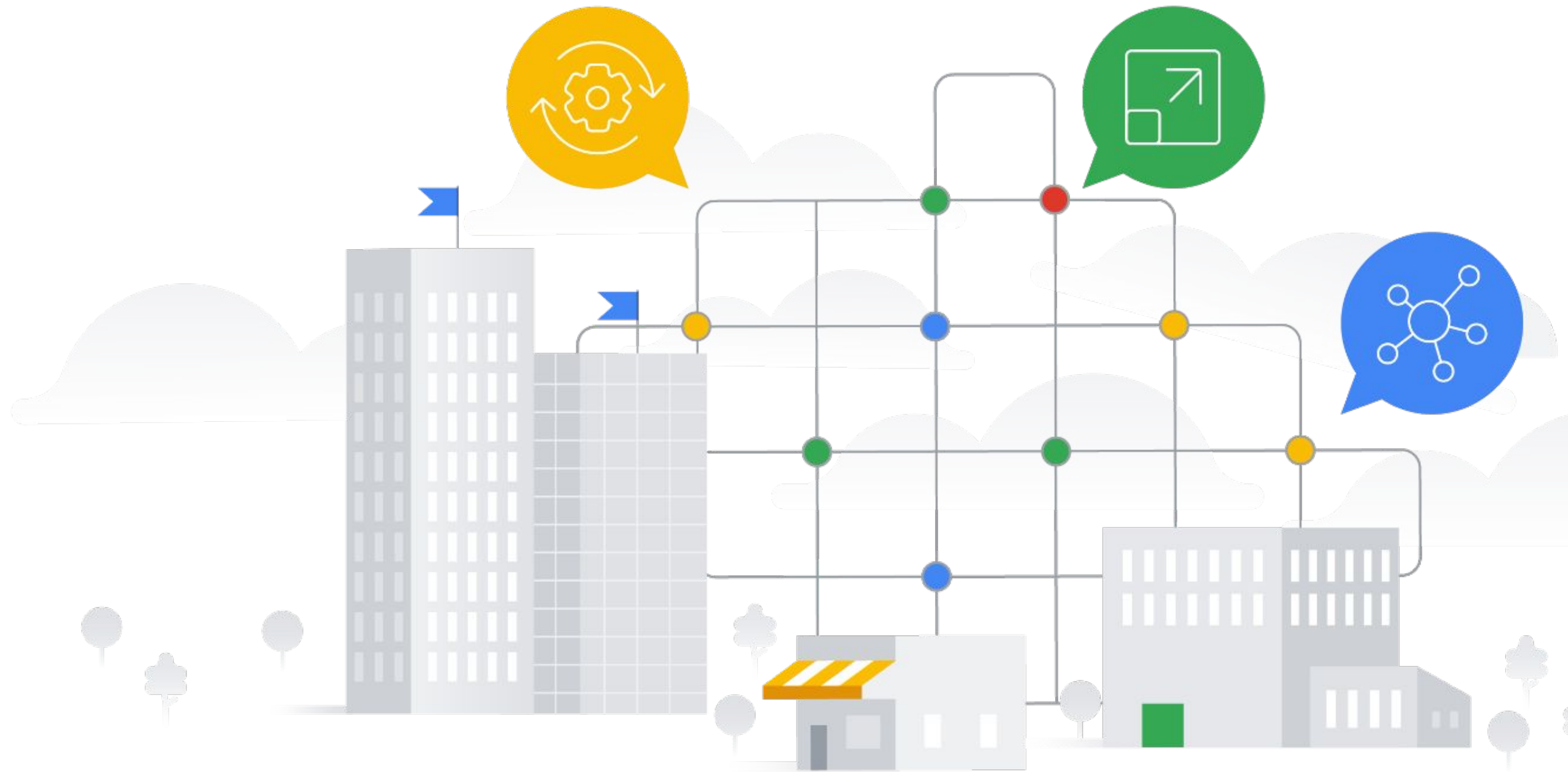


Lessons

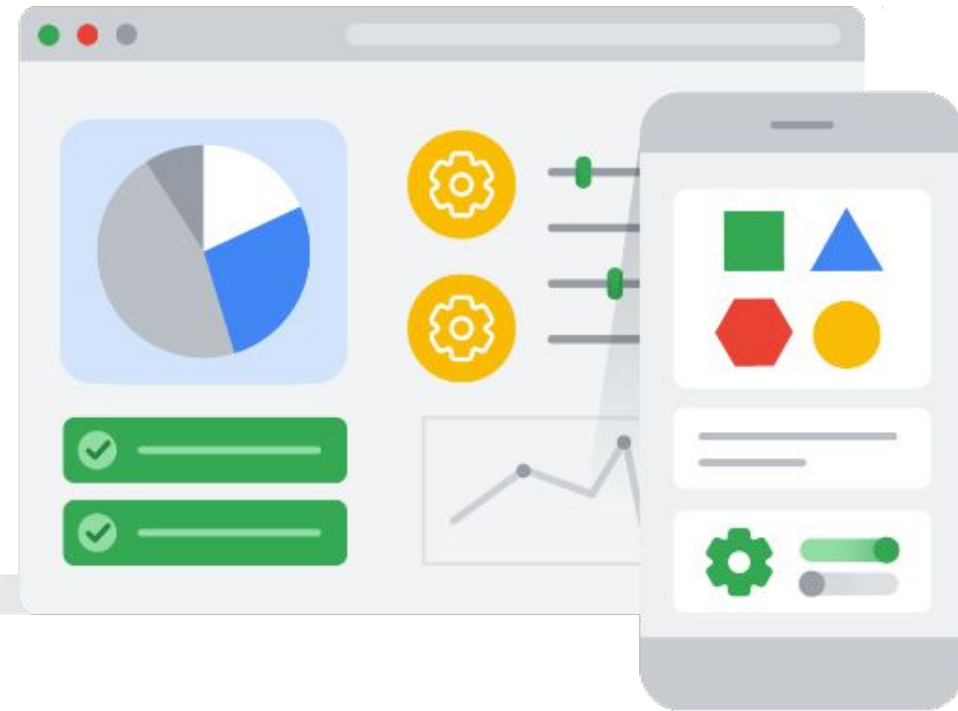
- 01 Financial governance and managing cloud costs
- 02 Operational excellence and reliability at scale
- 03 Sustainability with Google Cloud

Managing and scaling cloud resources effectively can be a complex task

Cloud operations refers to the set of practices and strategies employed to ensure the smooth functioning, optimization, and scalability of cloud-based systems.

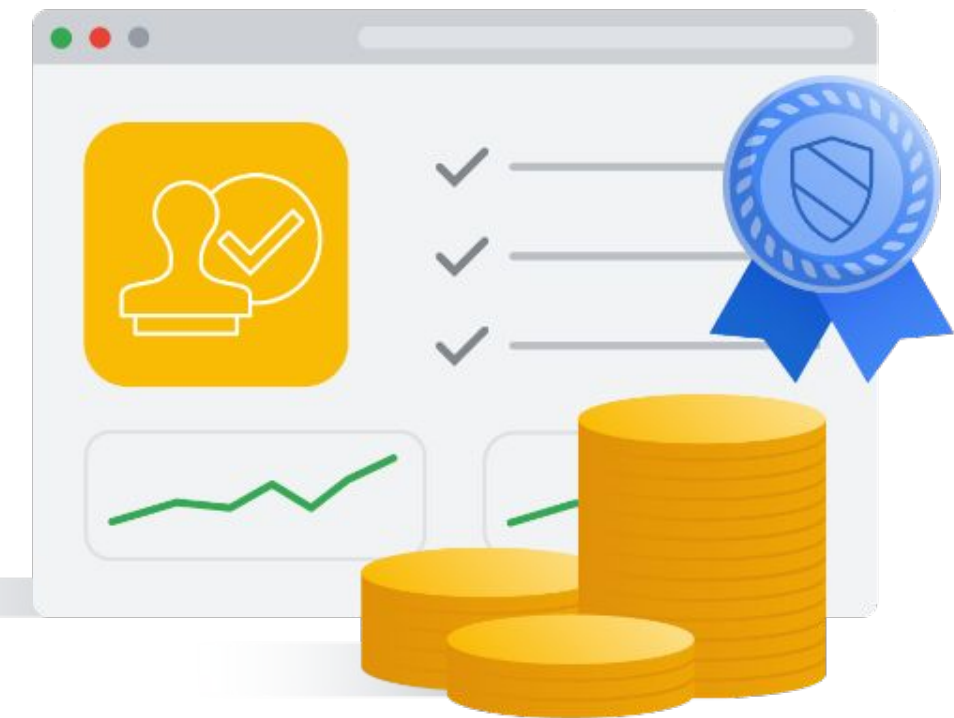


Cloud operations



Managing and monitoring:

- Infrastructure
- Applications
- Services



Adhering to best practices:

- Reliability
- Performance
- Security
- Cost optimization

Module 6

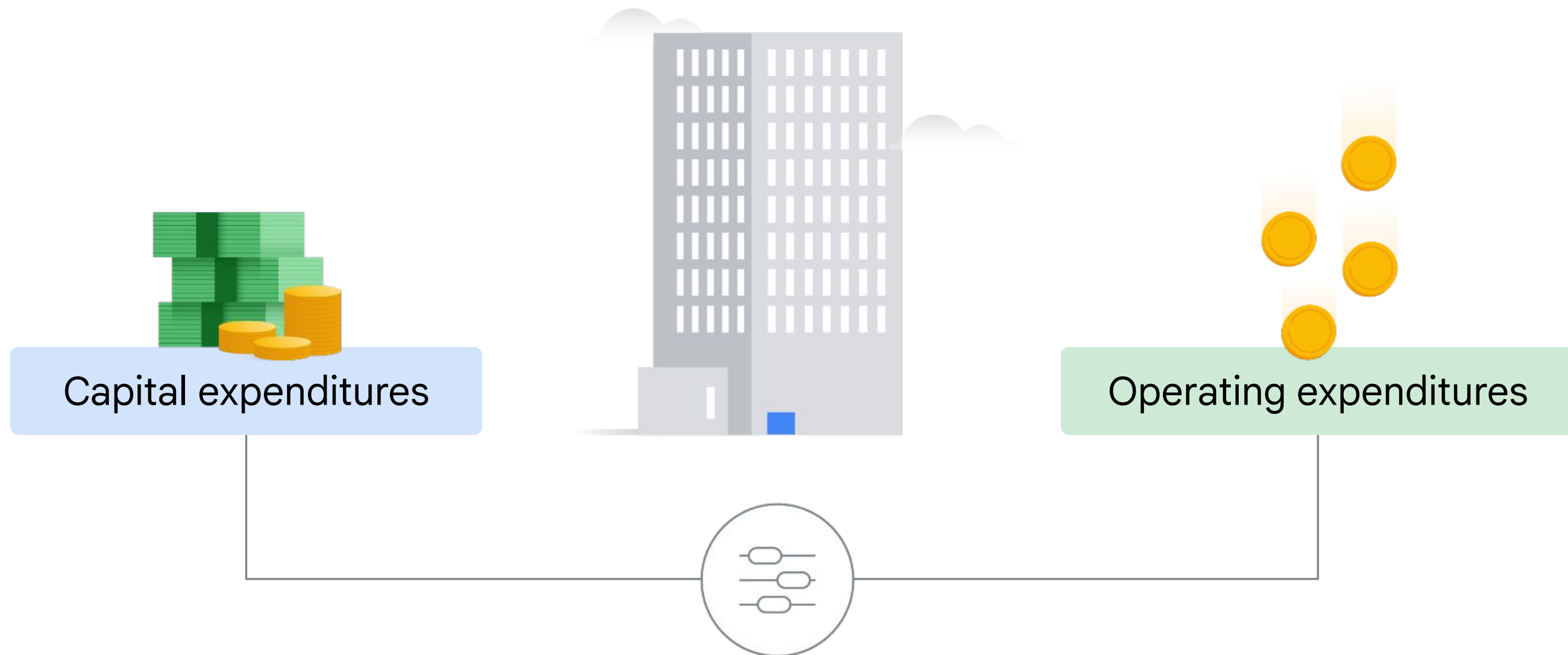
Scaling with Google Cloud Operations



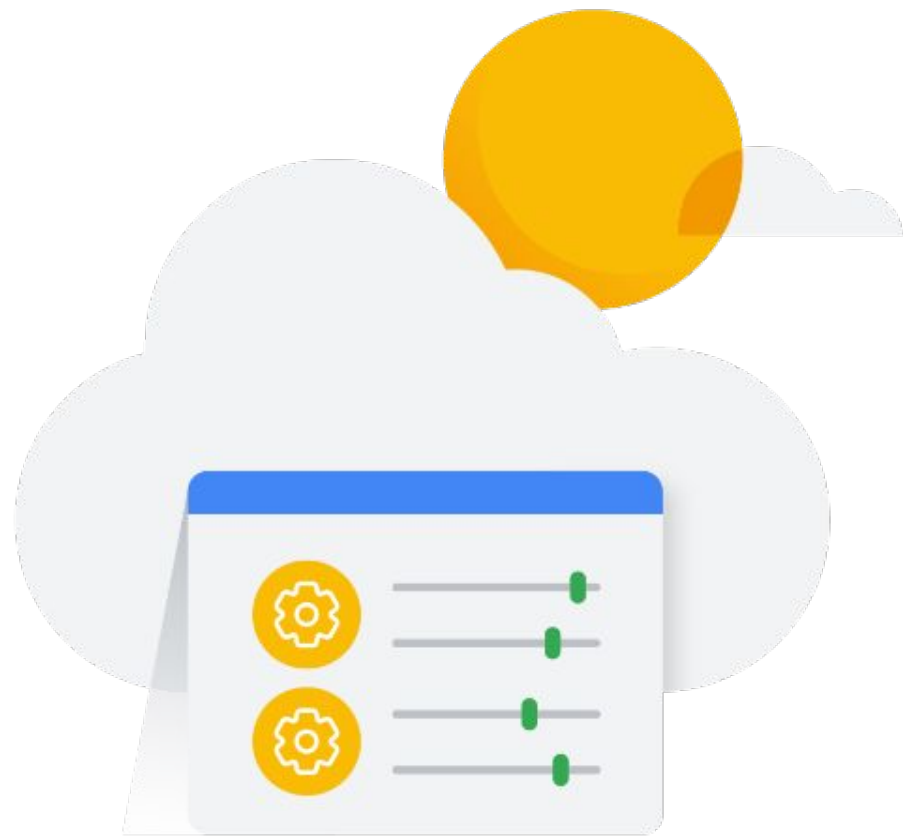
Lessons

- 01 Financial governance and managing cloud costs
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- 03 Sustainability with Google Cloud

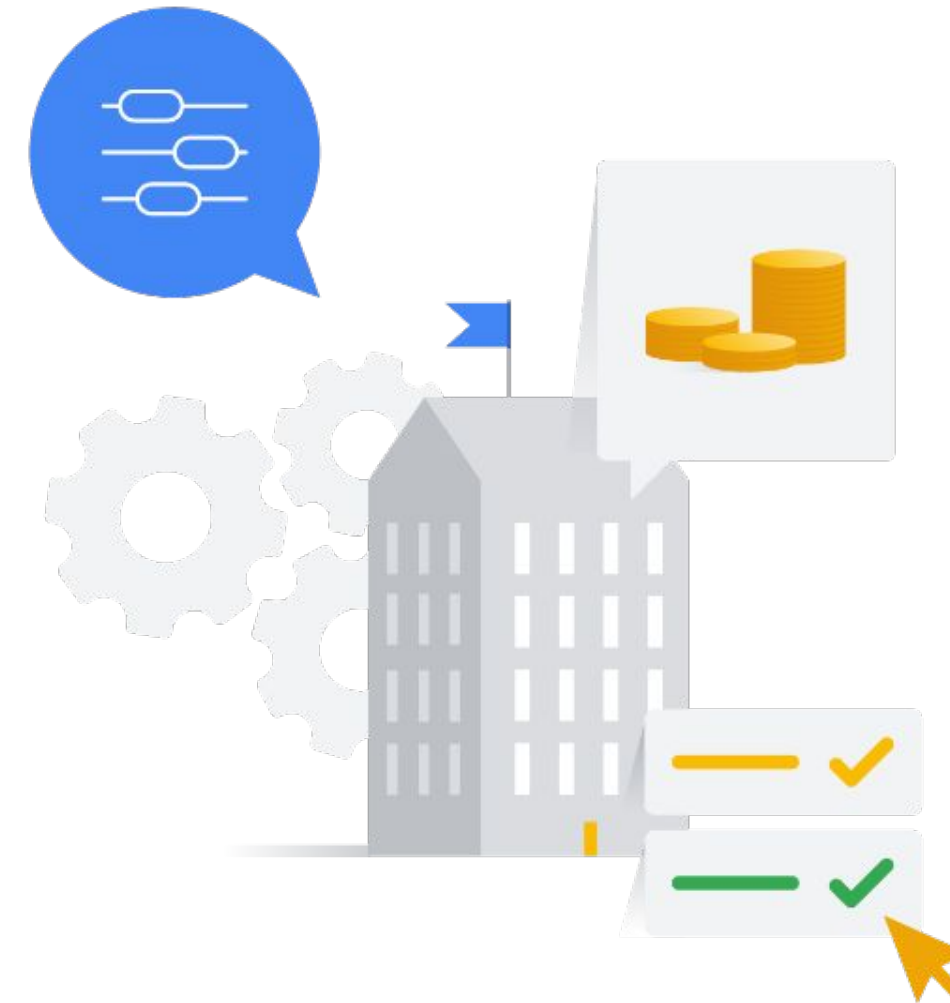
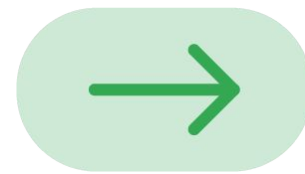
The transition CapEx to OpEx requires process and organizational changes



Cloud financial governance can mean the difference between peace of mind and spiraling costs



Precise, real-time control of what's being consumed



Cloud financial governance

Cloud costs impacts people, process, and technology

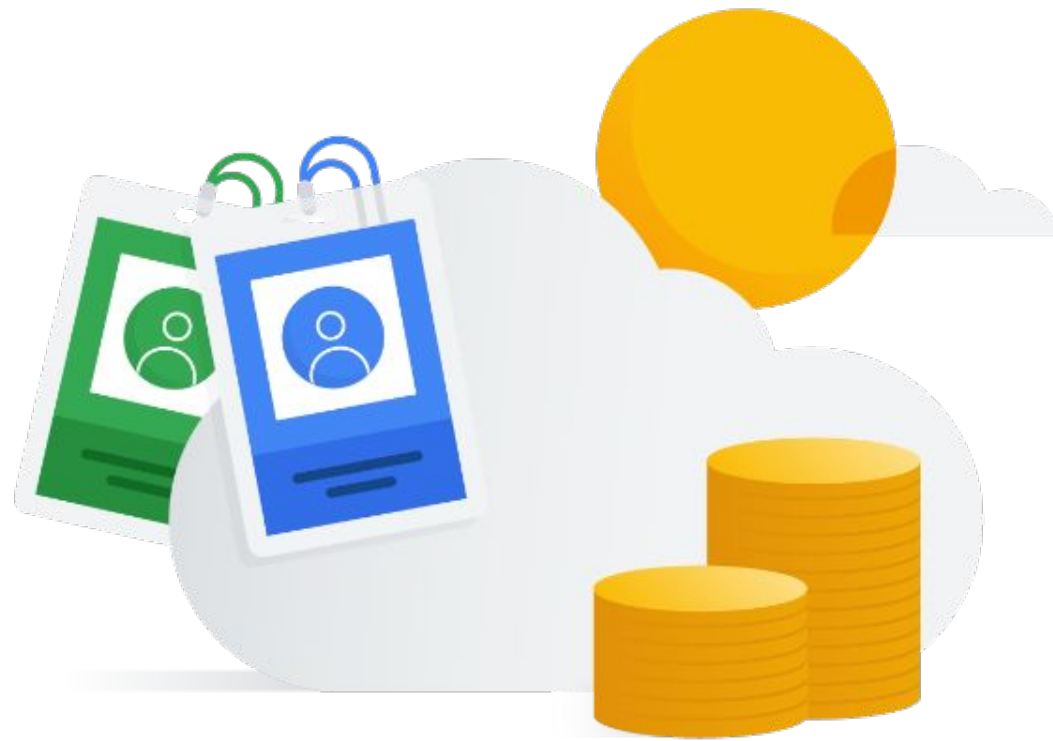


People

Process

Technology

People refers to the different roles involved in managing cloud costs



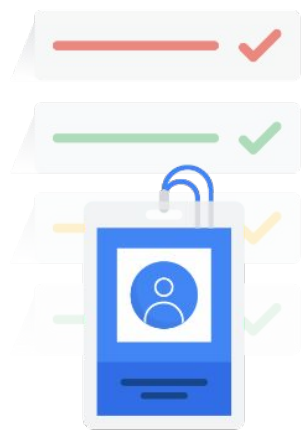
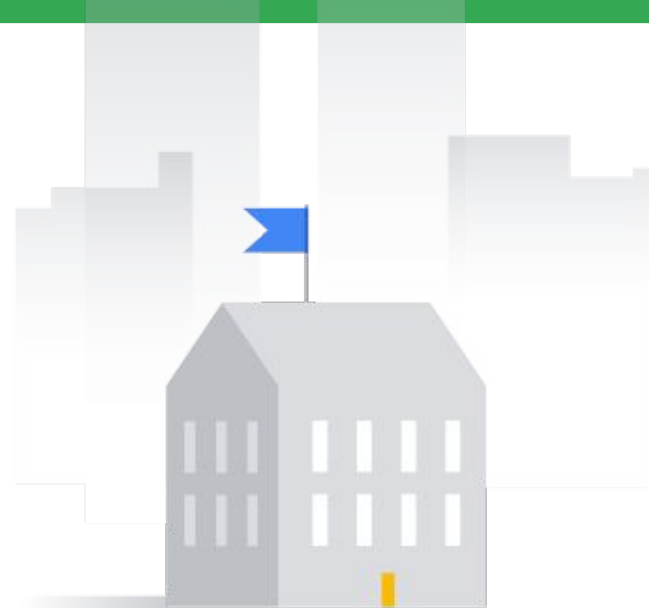
People

Process

Technology

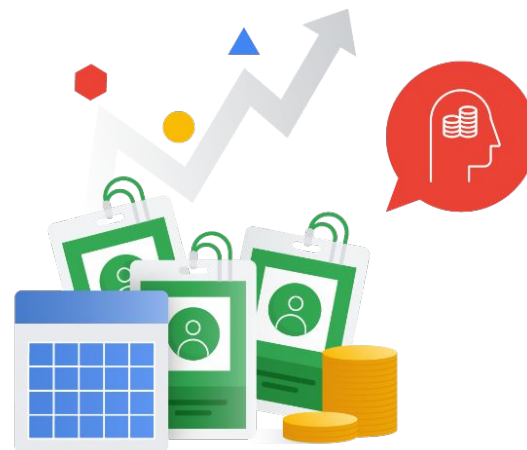
The roles people play at organizations of different sizes

Small organizations

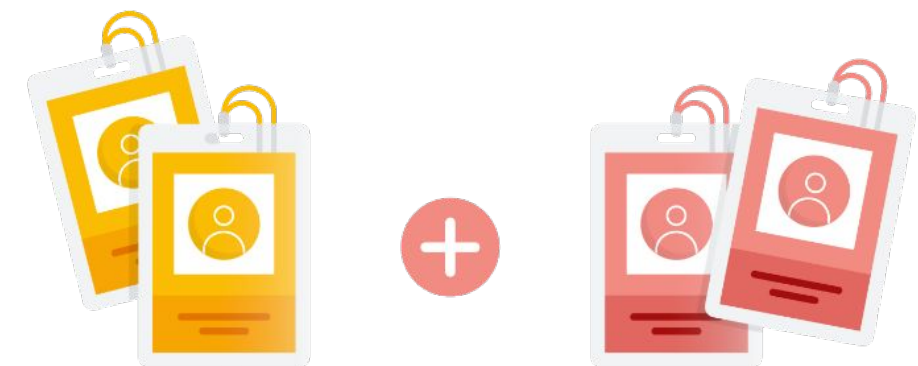
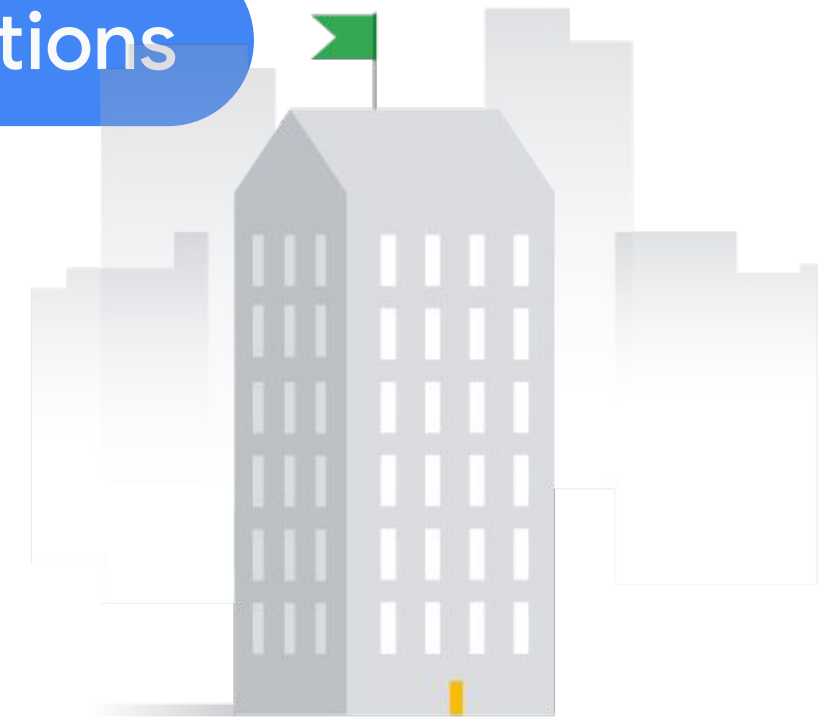


One person fulfills multiple roles.

Large organizations

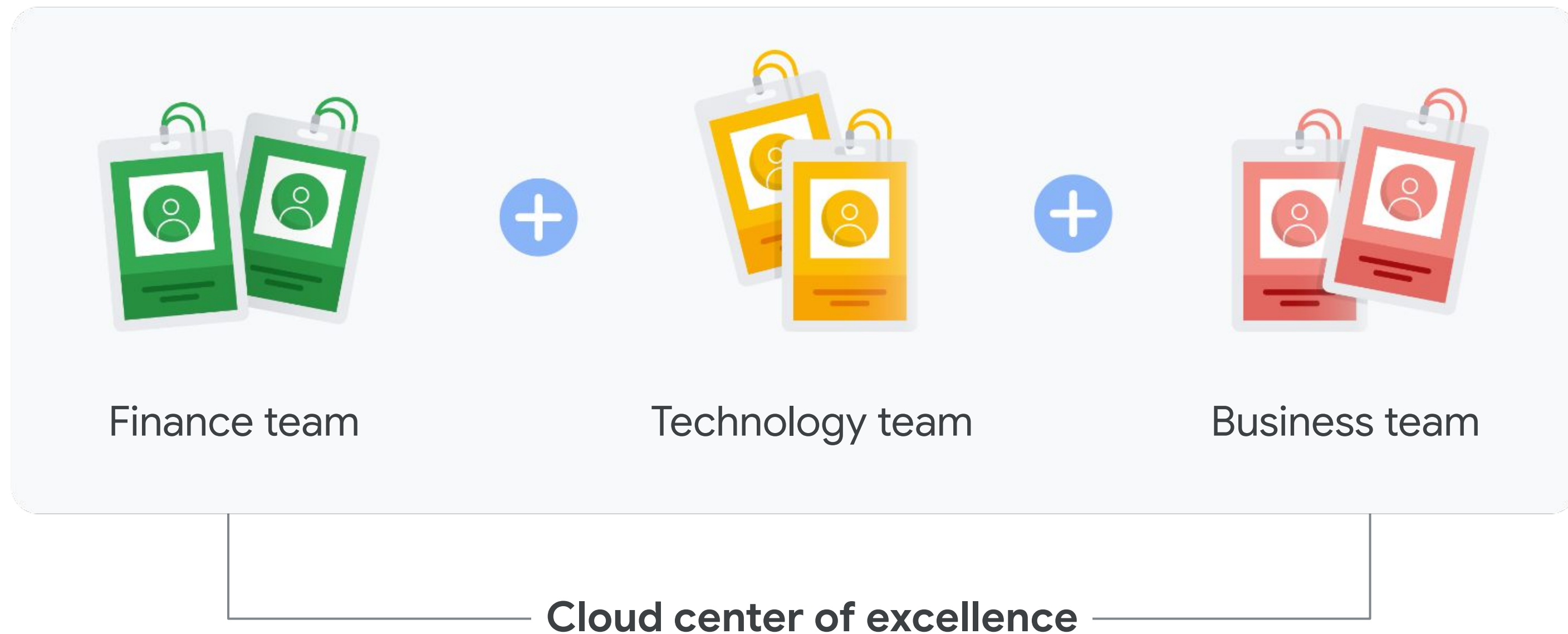


Finance team takes on financial planning.



Technology and business teams advise on cloud resource usage.

To manage cloud costs effectively, a partnership is required



Process

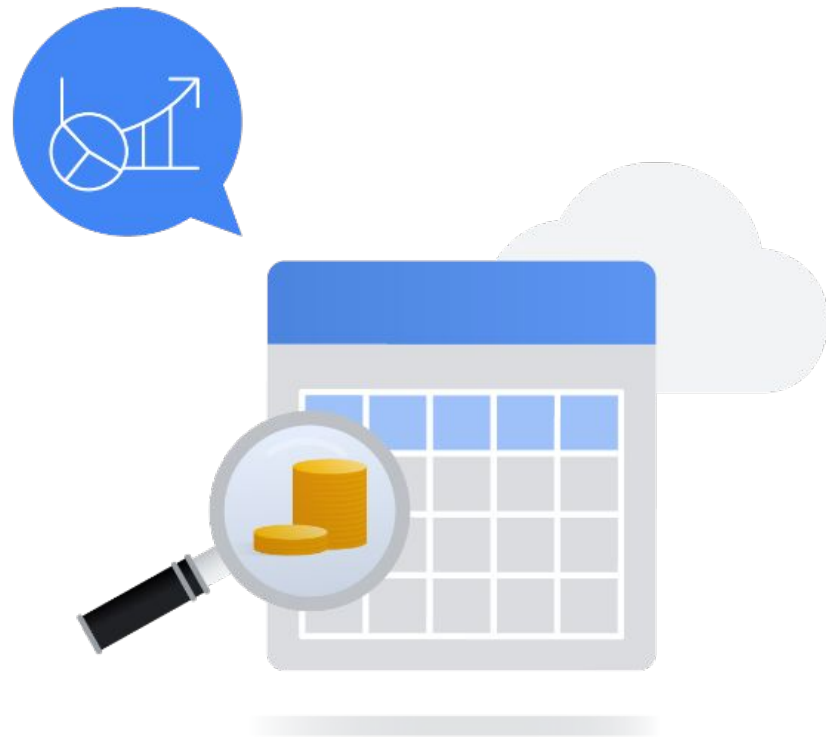


People

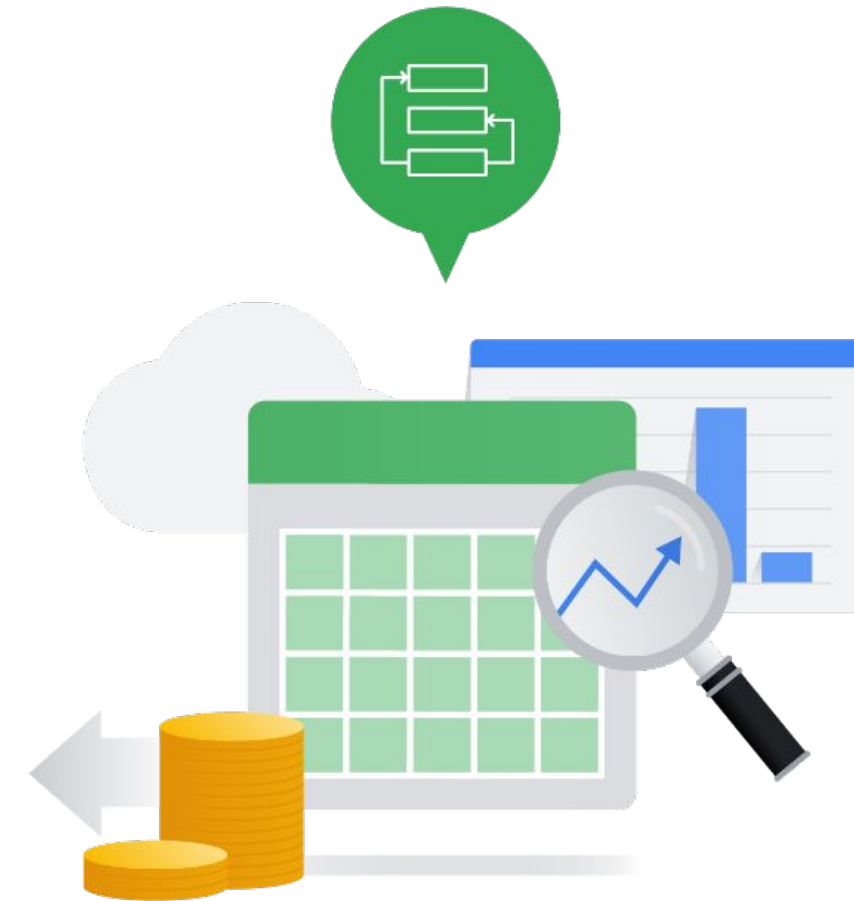
Process

Technology

Processes for organizations to implement



On a **daily or weekly** basis
Monitor and analyze cloud
usage and costs.



On a **weekly or monthly** basis
Analyze the results and charge back the
costs through the appropriate teams.
Make changes, if needed.

The importance of having a culture of accountability



Recognize waste, quickly act to eliminate it, and ensure they're maximizing their cloud investment.



Drive cross-group collaboration across technology, finance, and business teams to ensure their cloud spend aligns with broader business objectives.

Technology

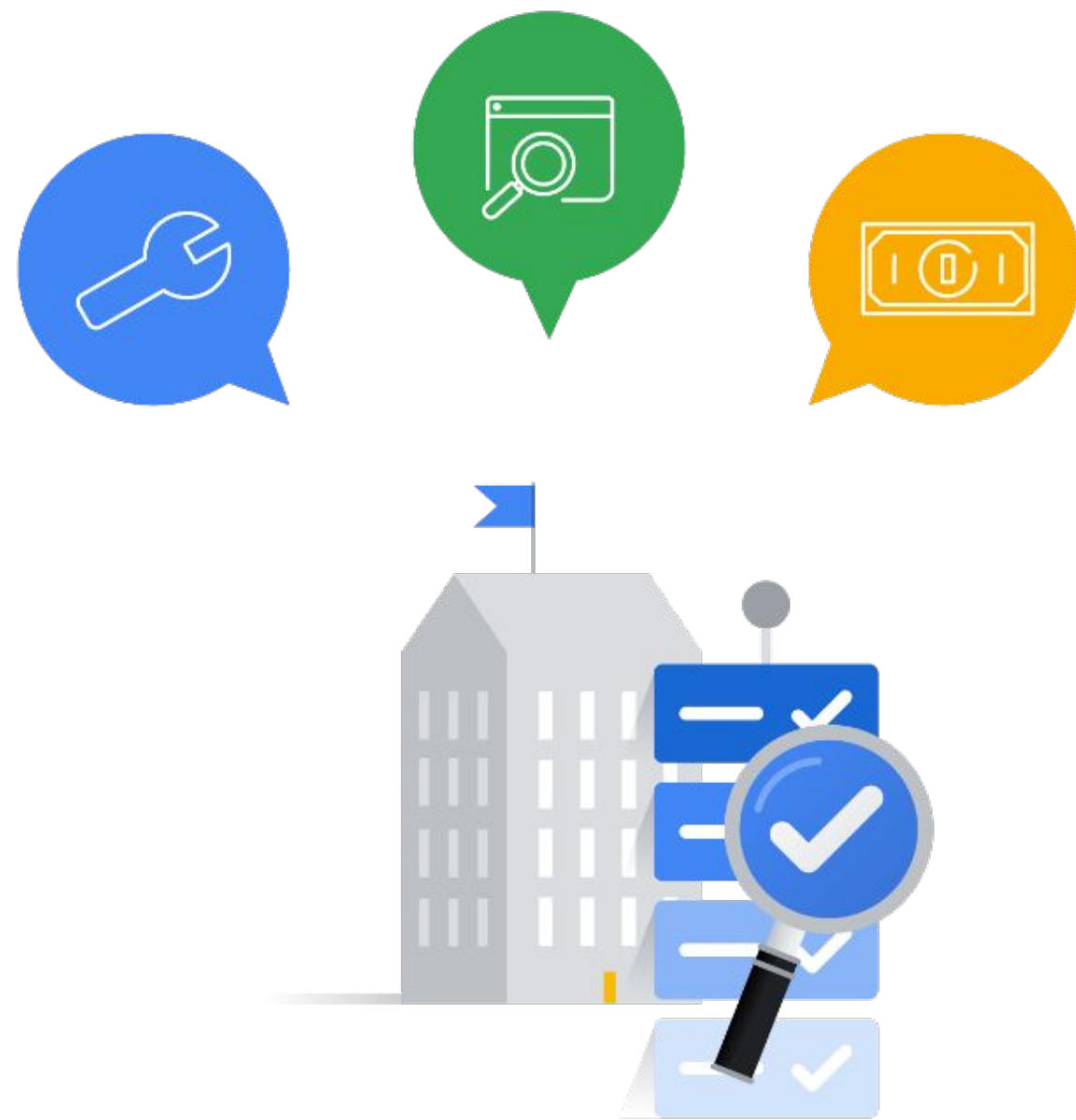


People

Process

Technology

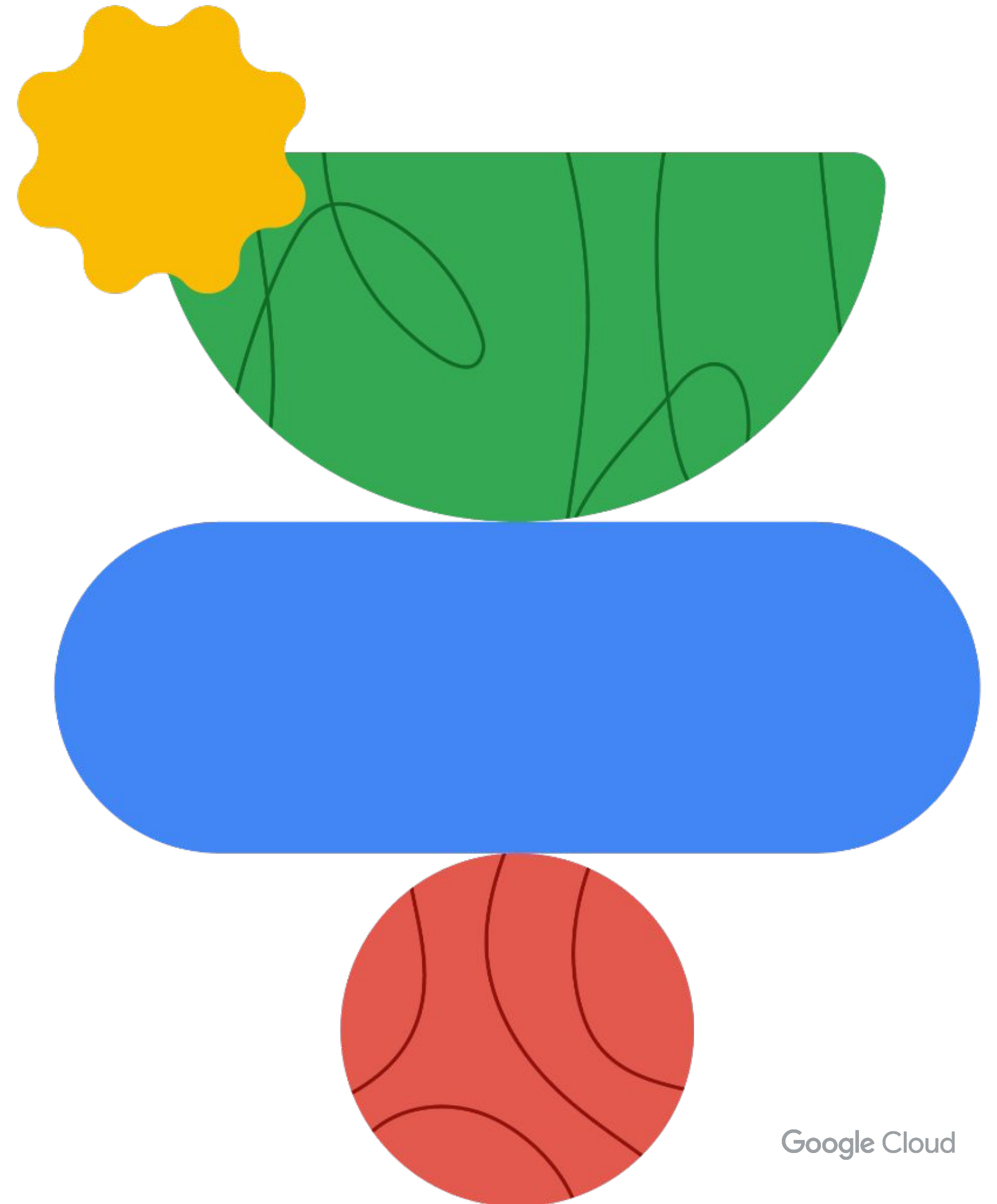
Google Cloud provides built-in tools to help organizations monitor and manage costs



These tools help organizations:

- Gain greater visibility
- Drive a culture of accountability for cloud spending across the organization
- Control costs to reduce risks of overspending
- Provide intelligent recommendations to optimize cost and usage

Cloud financial governance best practices



Cloud financial governance best practices



Identify who manages
cloud costs.



Understand invoices
versus cost management
tools.

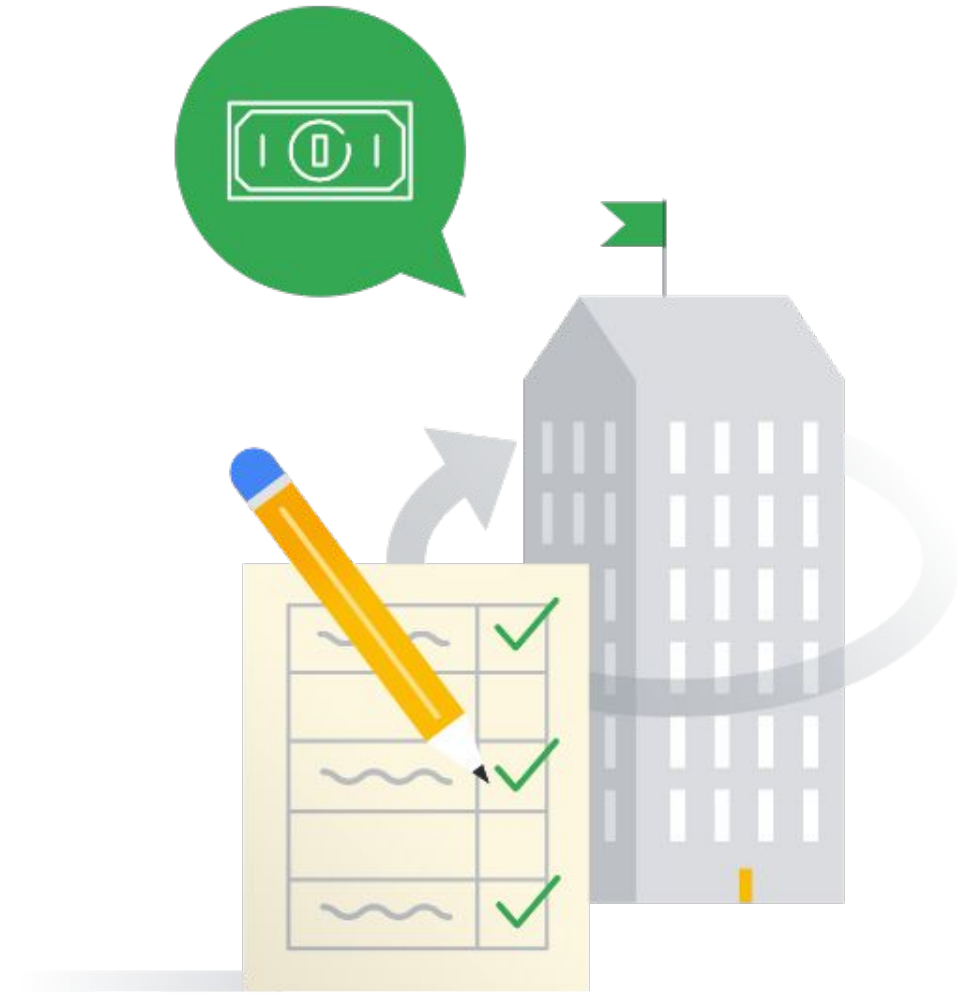
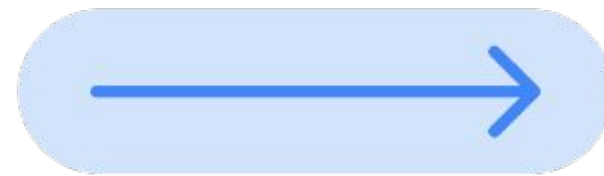


Use Google Cloud cost
management tools.

Establish a culture of accountability for costs across the organization



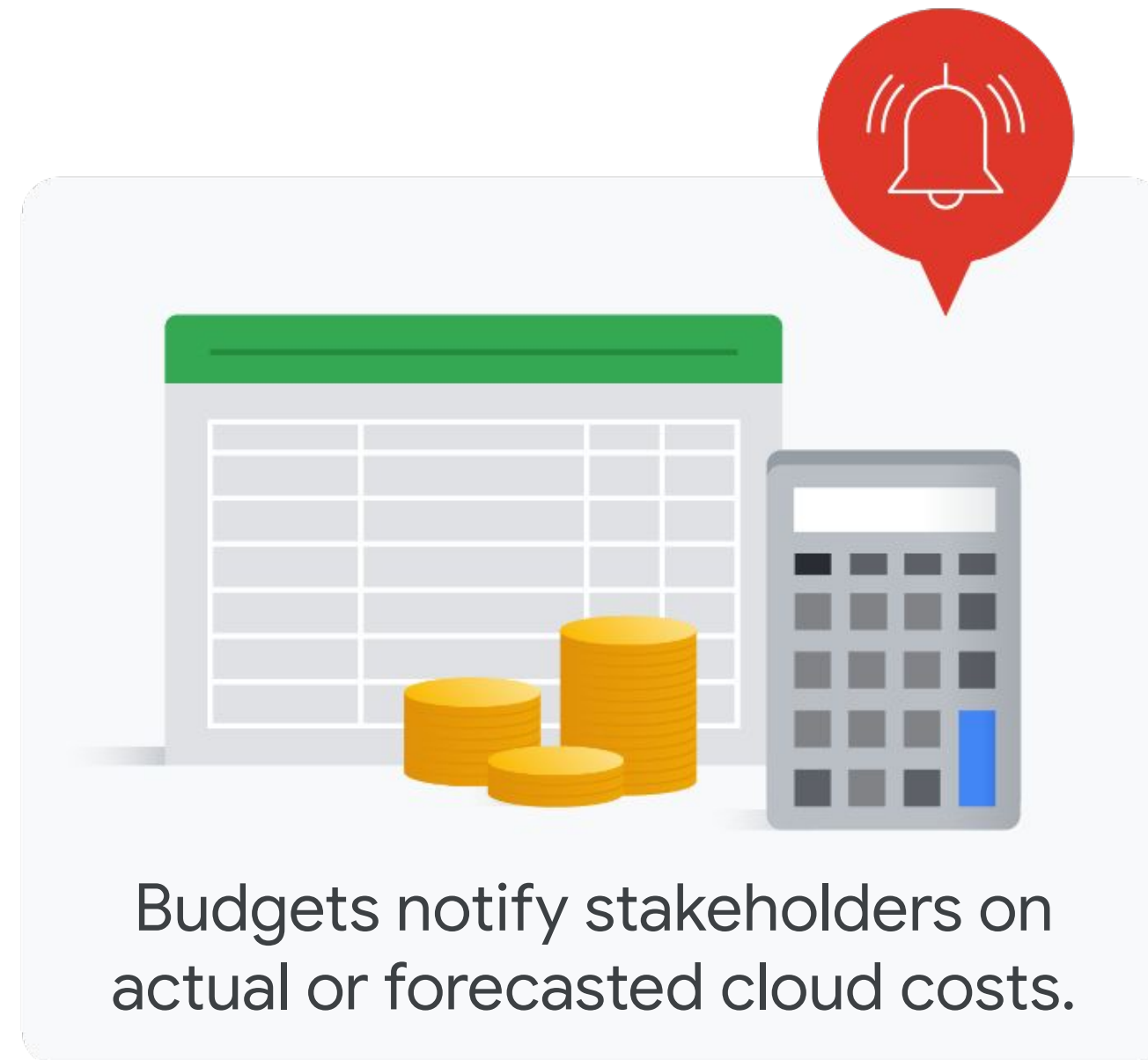
Cloud spending is decentralized and variable.



Define clear ownership for projects.

Share cost views with the departments and teams that are using cloud resources.

Policies and permissions help control who can spend and view costs across



Cloud financial governance best practices



Identify who manages
cloud costs.



Understand invoices
versus cost management
tools.



Use Google Cloud cost
management tools.

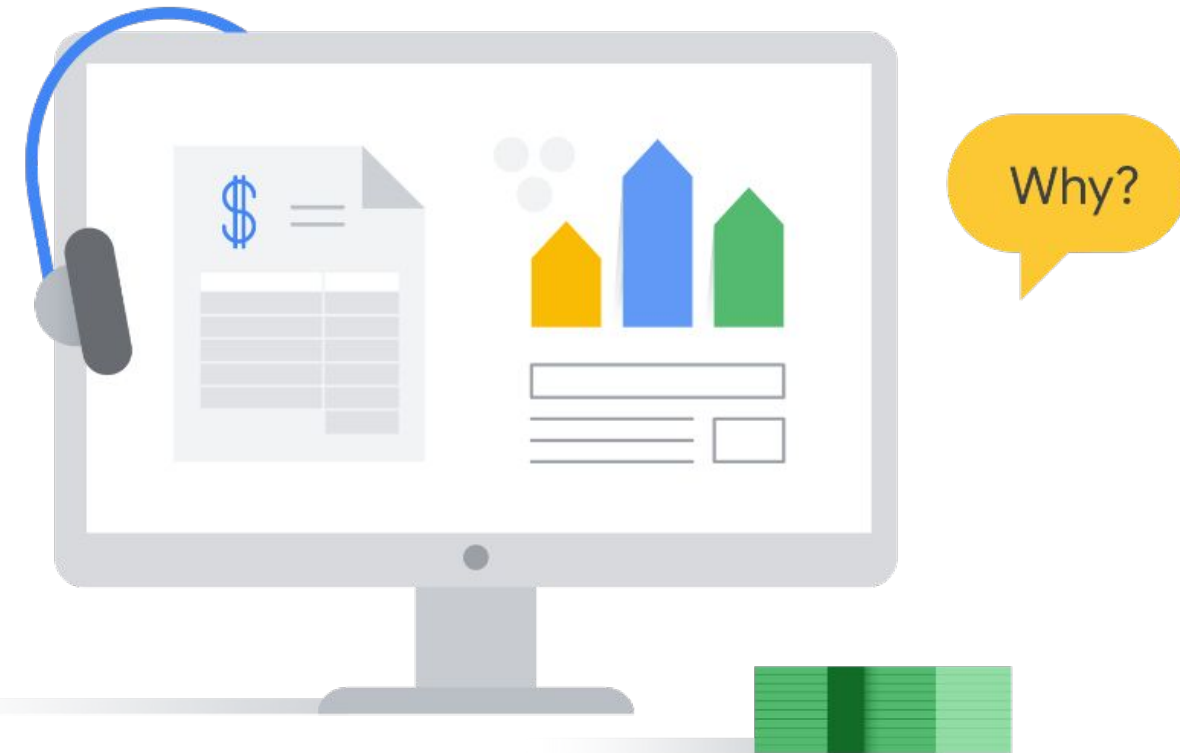
Invoices vs. cost management tools



Invoice

A document sent by a cloud service provider to a customer to request payment for the services that were used.

How much?



Cost management tool

Software to help track, analyze, and optimize cloud spend.

Cloud financial governance best practices



Identify who manages
cloud costs.

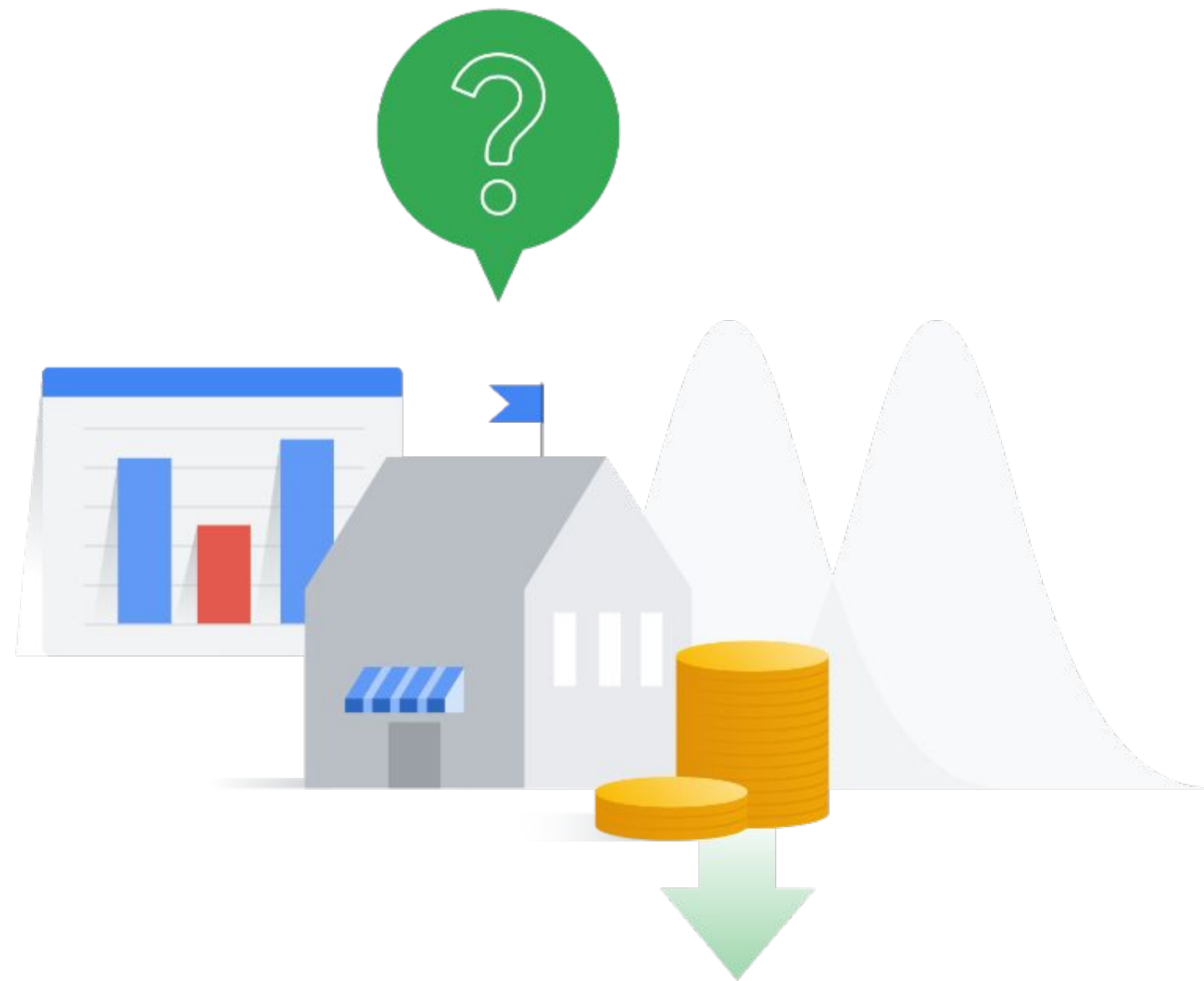


Understand invoices
versus cost management
tools.



Use Google Cloud cost
management tools.

Organizations must understand their cloud spend

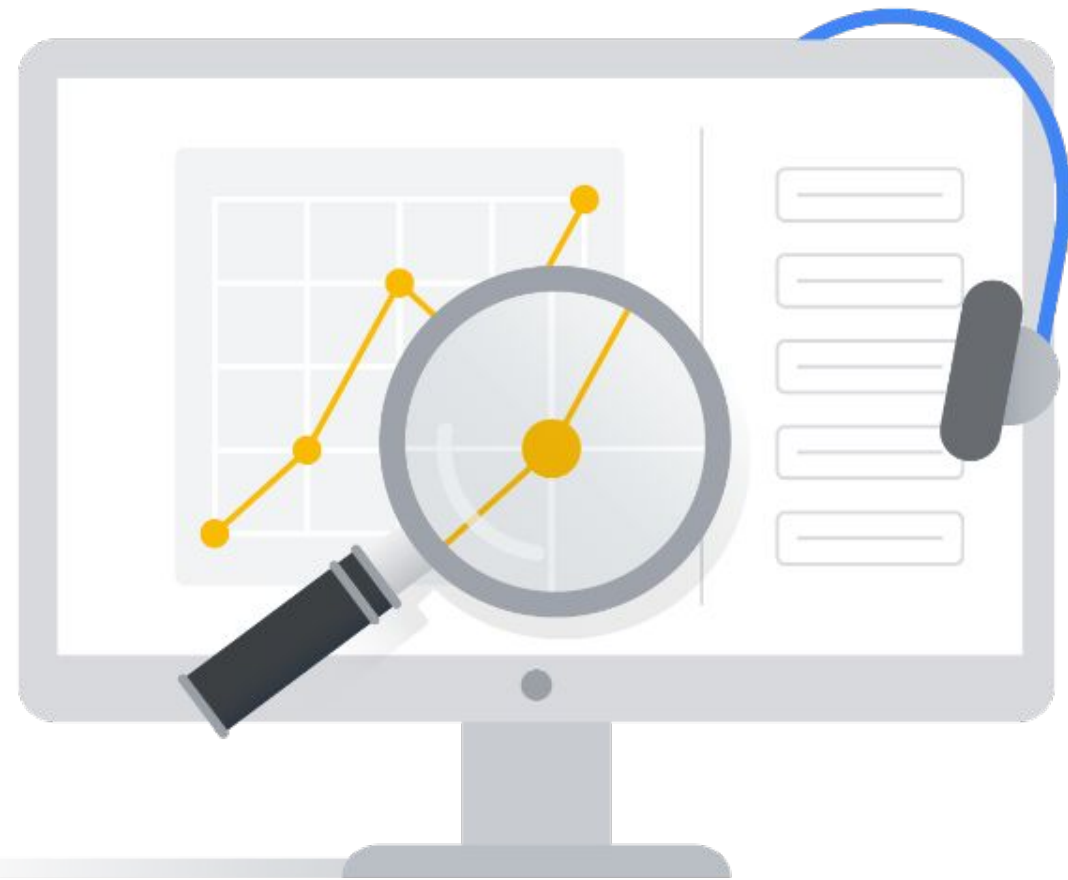


Capture what cloud resources are being used, by whom, for what purpose, and at what cost.

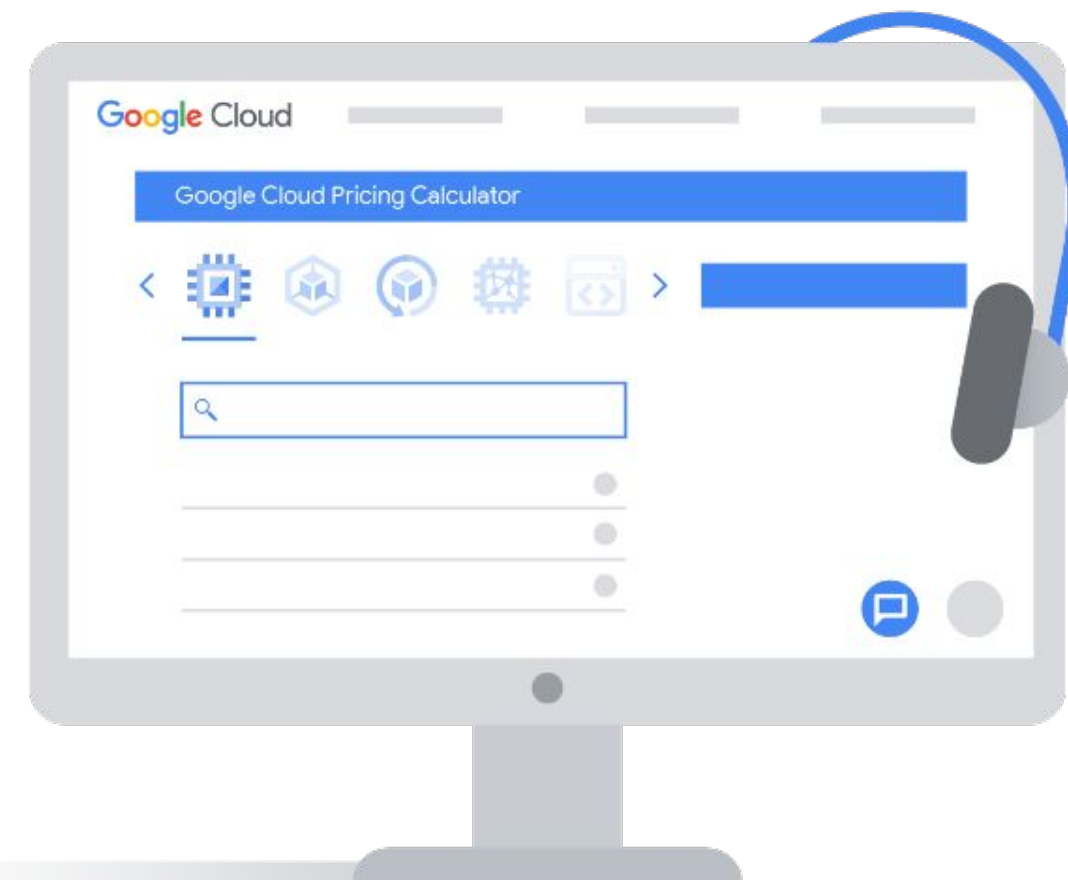
Determine who is responsible for monitoring that information, who is involved in managing costs, and how the spending information is reported on an ongoing basis.

Set up the cadence and format for ongoing communication with main cloud stakeholders.

Google Cloud helps organizations monitor cost trends and identify areas of waste



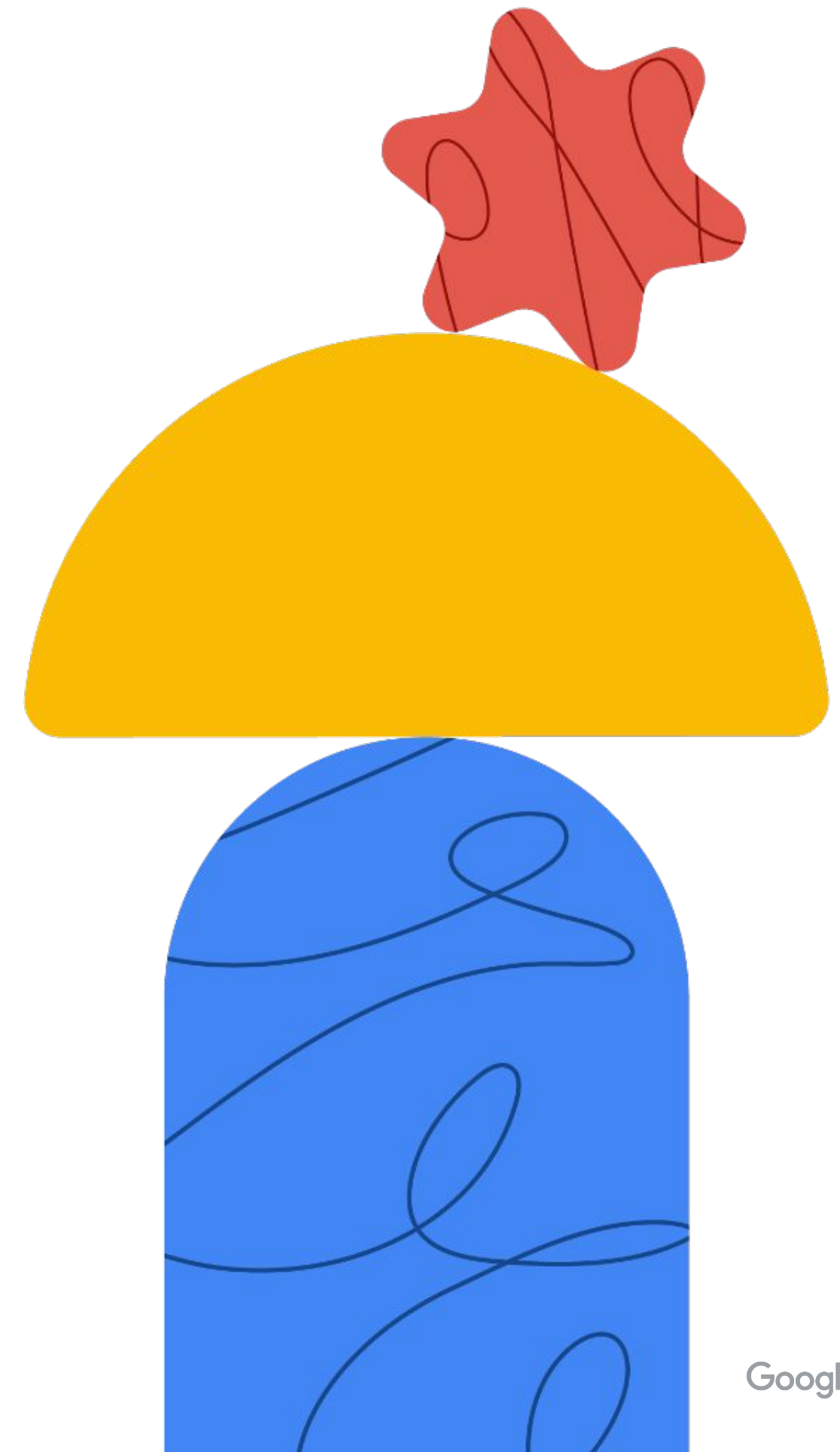
Built-in reporting capabilities



Google Cloud Pricing Calculator

cloud.google.com/products/calculator

Using the resource hierarchy to control access



The Google Cloud resource hierarchy

- It's a powerful tool that can be used to control access to cloud resources.
- This tree-like structure organizes resources into logical groups.

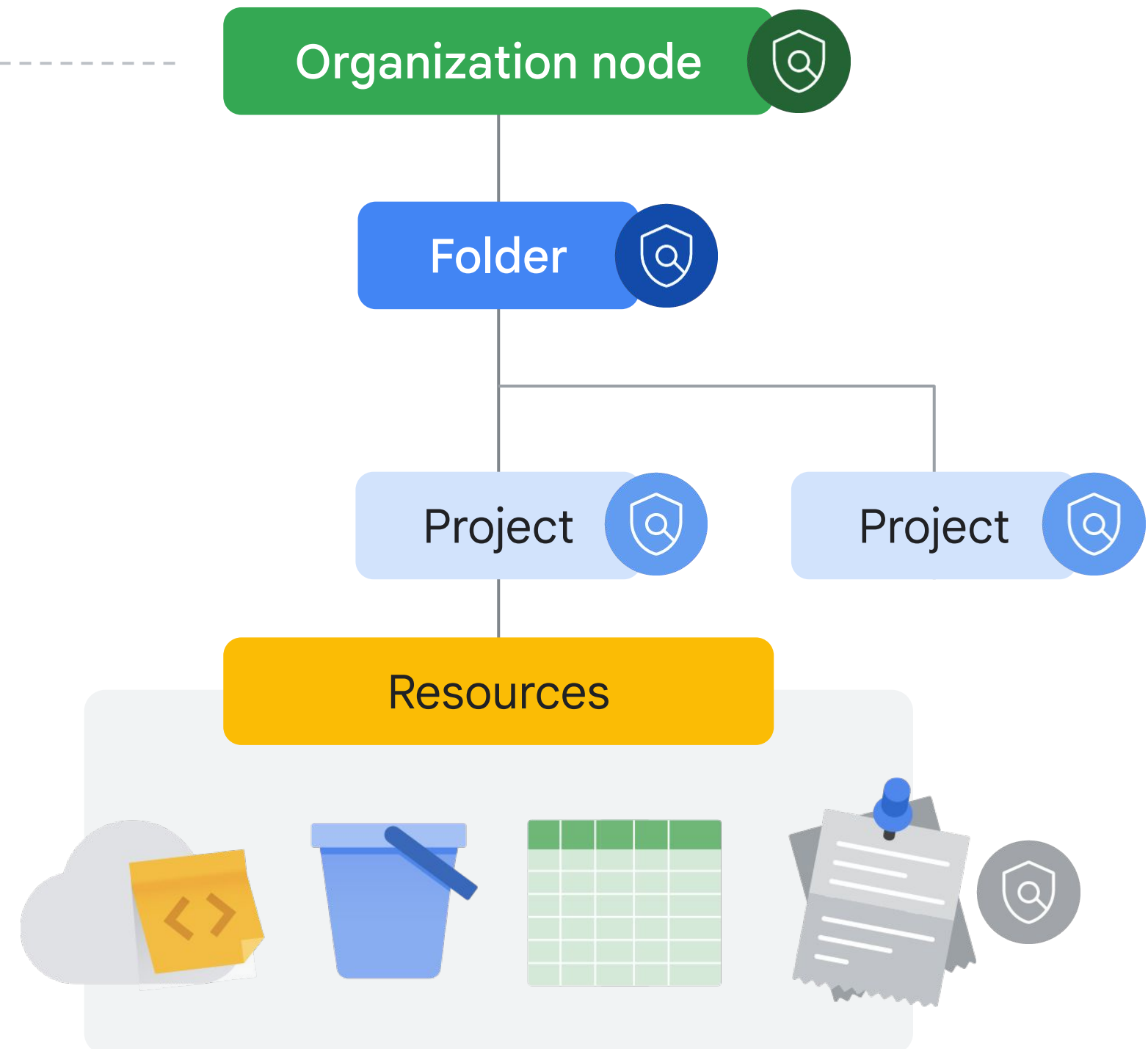


Policies

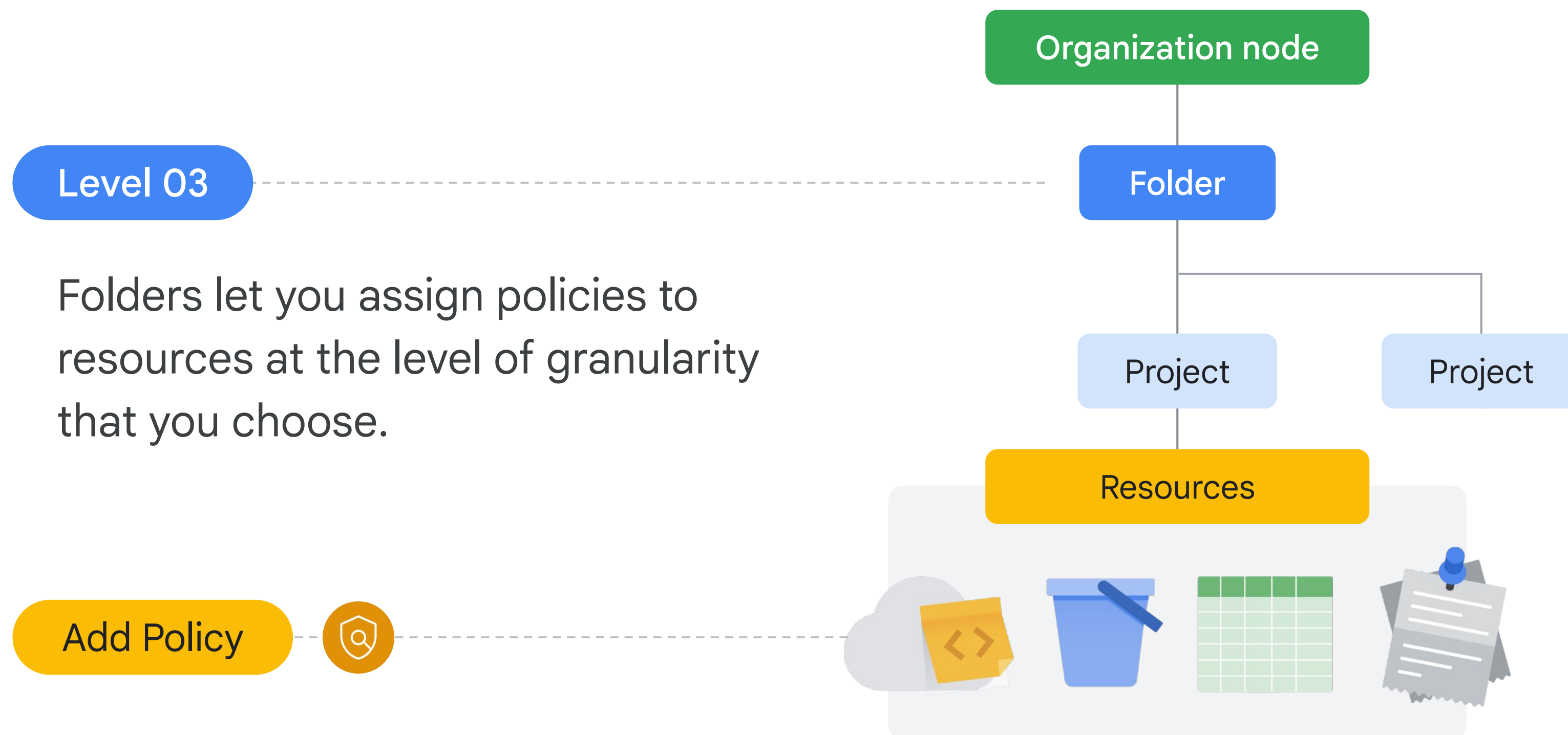
Add Policy



A set of rules that define who can access a resource and what they can do with it.

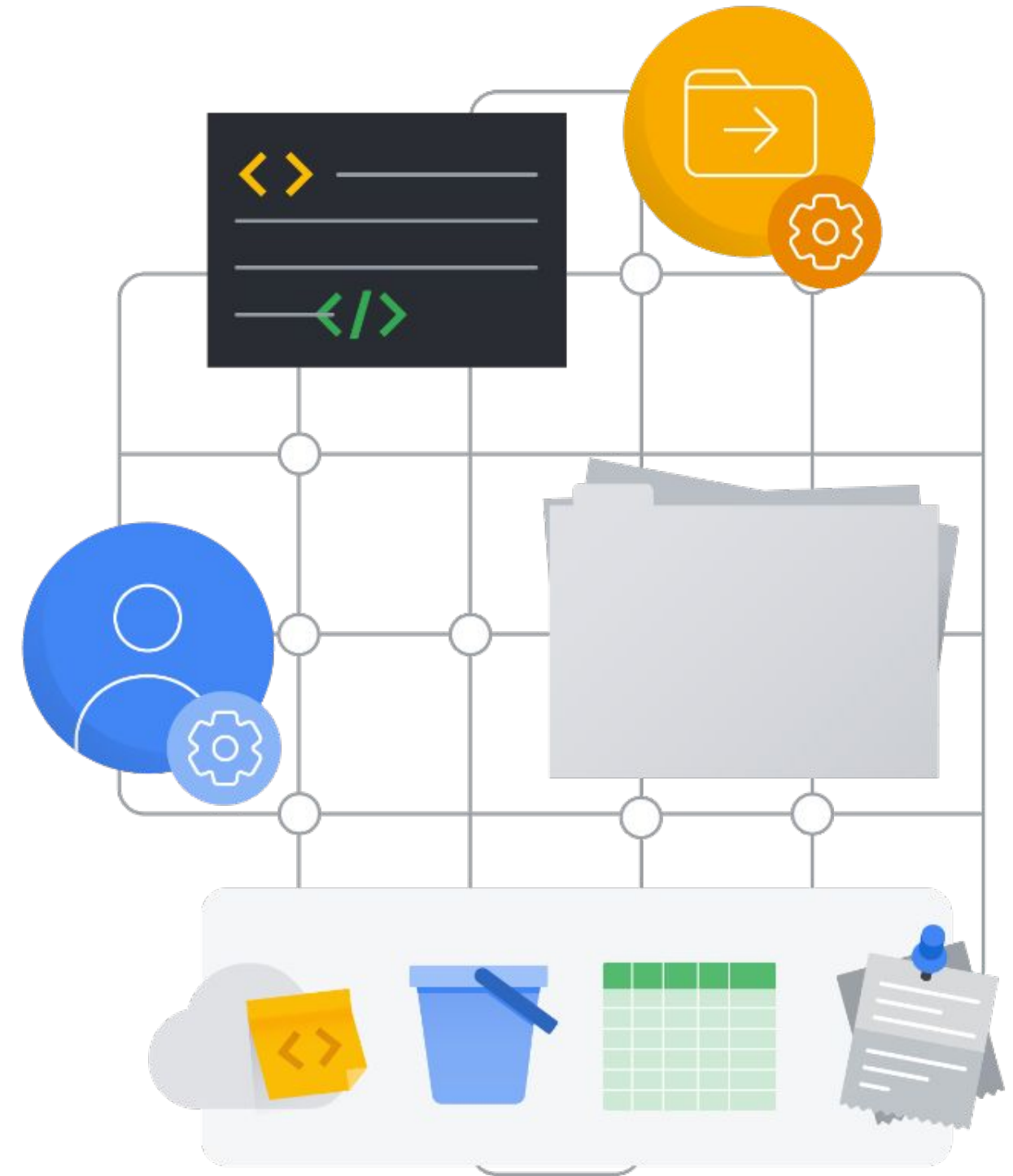


Folders

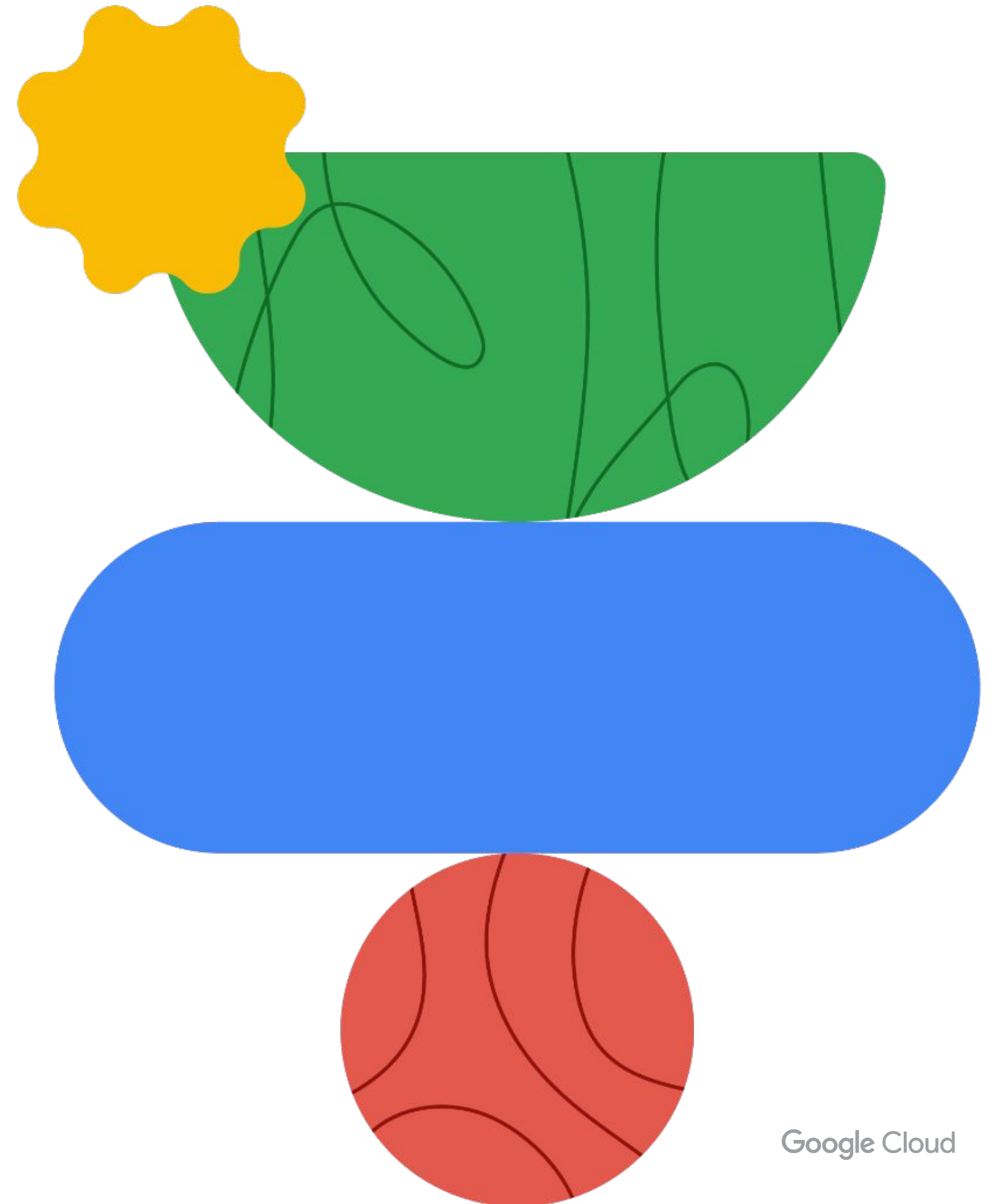


Benefits of using the resource hierarchy to control access

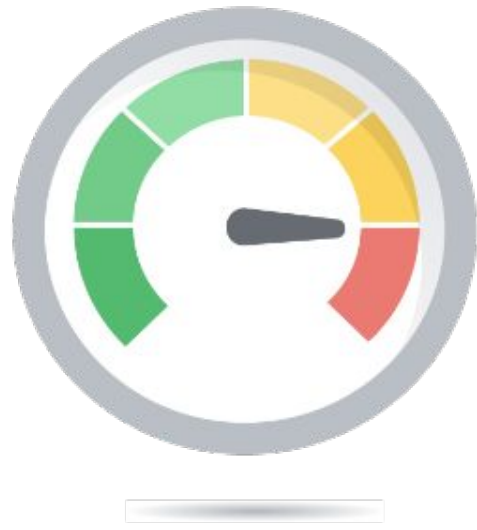
- 01 Granular access control
- 02 Inheritance and propagation rules
- 03 Security and compliance
- 04 Strong visibility and auditing capabilities



Controlling cloud consumption

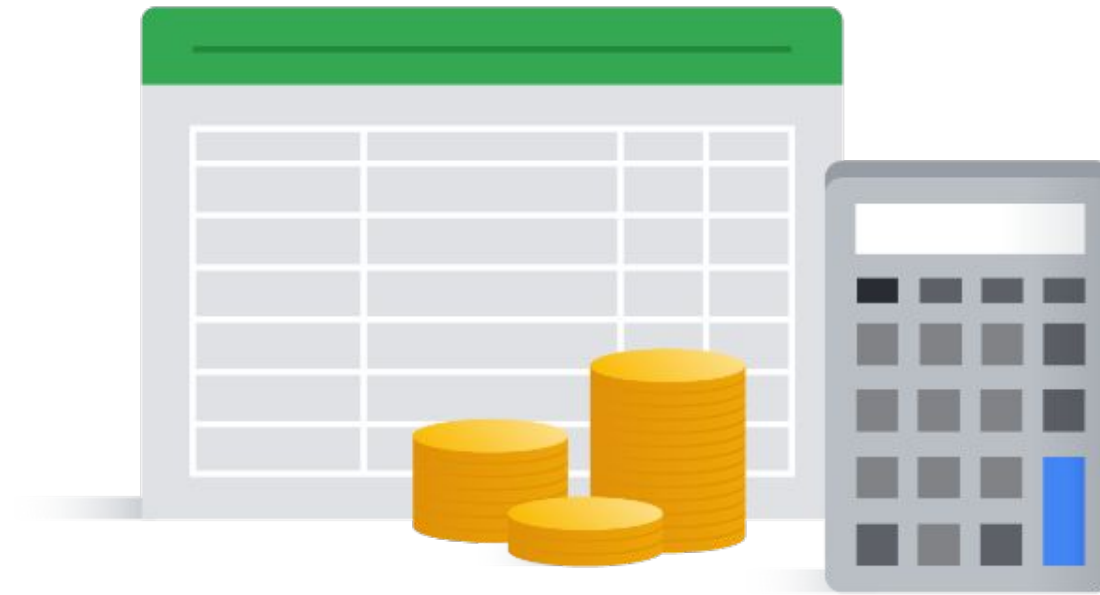


Google Cloud tools to help control cloud consumption



Resource quota policies

Set **limits** on the amount of resources that can be used by a project or user.



Budget threshold rules

Set **alerts** to be informed when your cloud costs exceed a certain threshold.



Cloud Billing reports

A **reactive method** to help track and understand what's already been spent.

Optimizing costs through committed use discounts



Committed use discounts (CUDs)

Discounted prices in exchange for your commitment to use a minimum level of resources for a specific term.

Quiz

Question

Which represents the lowest level in the Google Cloud resource hierarchy?

- A. Folders
- B. Projects
- C. Organization node
- D. Resources

Quiz

Answer

Which represents the lowest level in the Google Cloud resource hierarchy?

- A. Folders
- B. Projects
- C. Organization node
- D. Resources



Quiz

Question

Which feature lets you set limits on the amount of resources that can be used by a project or user?

- A. Quota policies
- B. Billing reports
- C. Budget alerts
- D. Committed use discounts

Quiz

Answer

Which feature lets you set limits on the amount of resources that can be used by a project or user?

- A. Quota policies
- B. Billing reports
- C. Budget alerts
- D. Committed use discounts



Module 6

Scaling with Google Cloud Operations



Lessons

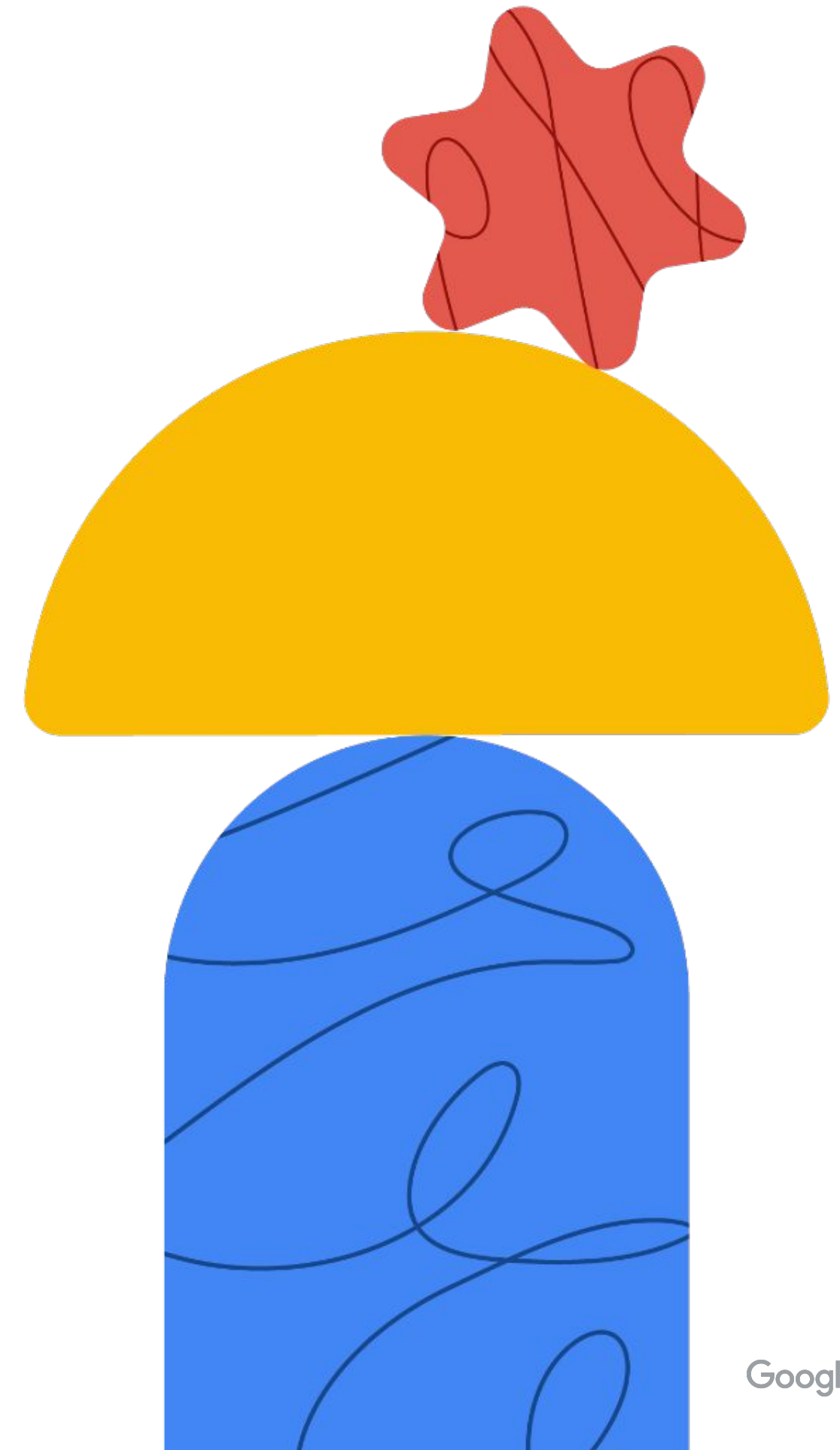
- 01 Financial governance and managing cloud costs
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Operational excellence and reliability

The ability to optimize operations and ensure uninterrupted service delivery, even when handling increasing workloads and complexities in the cloud.



The fundamentals of cloud reliability



Developers vs. operators

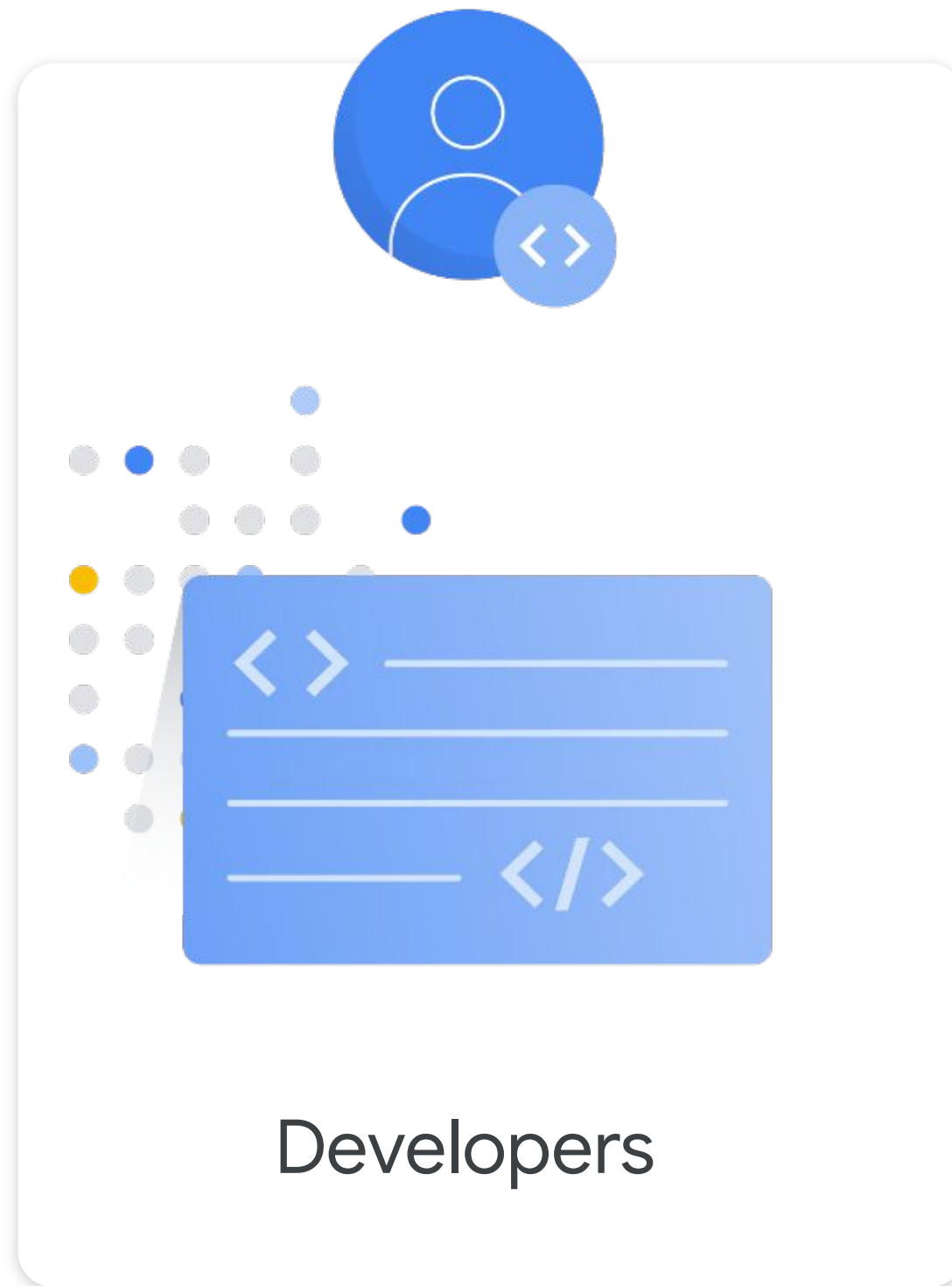


Developers write code for systems and applications.



Operators ensure those systems and applications operate reliably.

Expectations of developers



- Are expected to be agile.
- Are often pushed to write and deploy code quickly.
- Aim to release new functions frequently.
- Increase core business value with new features.
- Release fixes fast.

Expectations of operators

- Are expected to keep the system stable.
- Often prefer to work more slowly to ensure reliability and consistency.



Operators

It can be difficult for either group to identify the source of problems and resolve them quickly



Traditionally, developers pushed their code to operators who often had little understanding of how the code runs in a live environment.

DevOps emphasizes collaboration and communication between development and operations teams



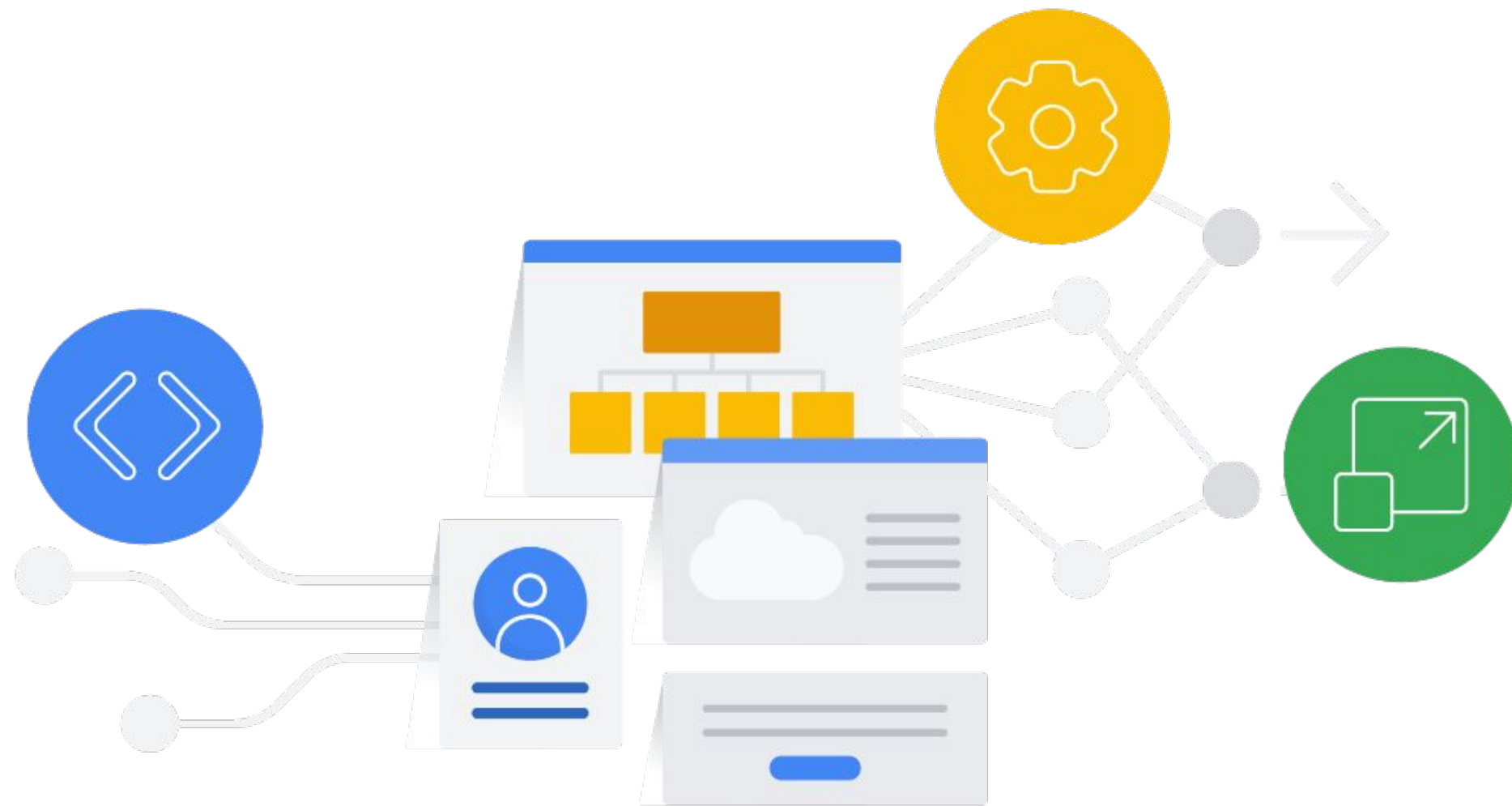
Developers



Operators

- ✓ Shared responsibility
- ✓ Automation
- ✓ Continuous improvement

Site Reliability Engineering (SRE)



Ensures the reliability, availability, and efficiency of software systems and services deployed in the cloud.

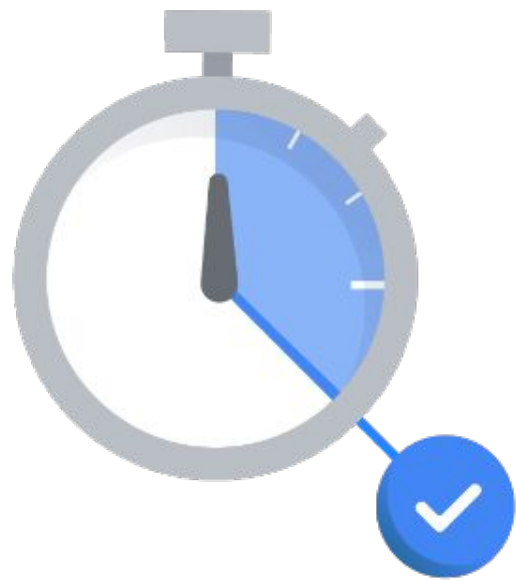
Monitoring is the foundation of product reliability



Monitoring

- ✓ Reveals what needs urgent attention.
- ✓ Shows trends in application usage patterns.
- ✓ Can yield better capacity planning.
- ✓ Help improve an application client's experience.

Four golden signals measure a system's performance and reliability



01

Latency



02

Traffic



03

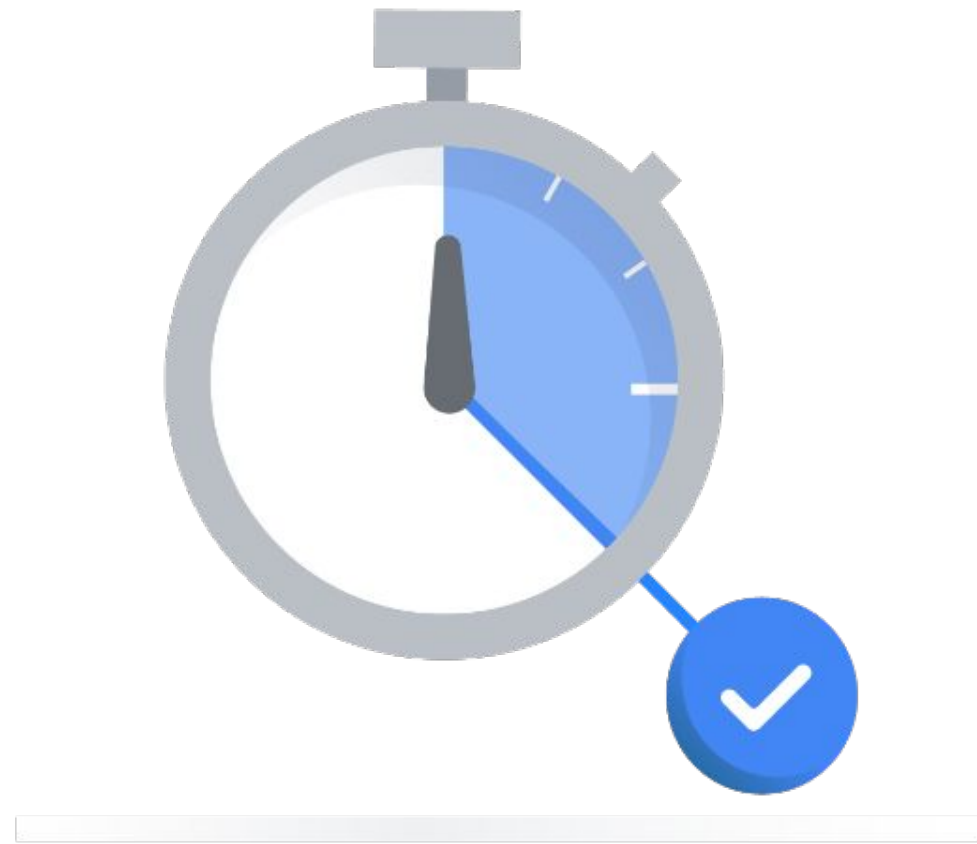
Saturation



04

Errors

Latency



Latency measures how long it takes for a particular part of a system to return a result.

- 01 It directly affects the user experience.
- 02 Changes in latency could indicate emerging issues.
- 03 Its values might be tied to capacity demands.
- 04 It can be used to measure system improvements.

Traffic



Traffic measures how many requests reach your system.

01 It's an indicator of current system demand.

02 Its historical trends are used for capacity planning.

03 It's a core measure when calculating infrastructure spend.

Saturation



Saturation measures how close to capacity a system is.

- 01 It's an indicator of how full the service is.
- 02 It focuses on the most constrained resources.
- 03 It's frequently tied to degrading performance as capacity is reached.

Errors



Errors are events that measure system failures or other issues.

01

They may indicate that something is failing.

02

They may indicate configuration or capacity issues.

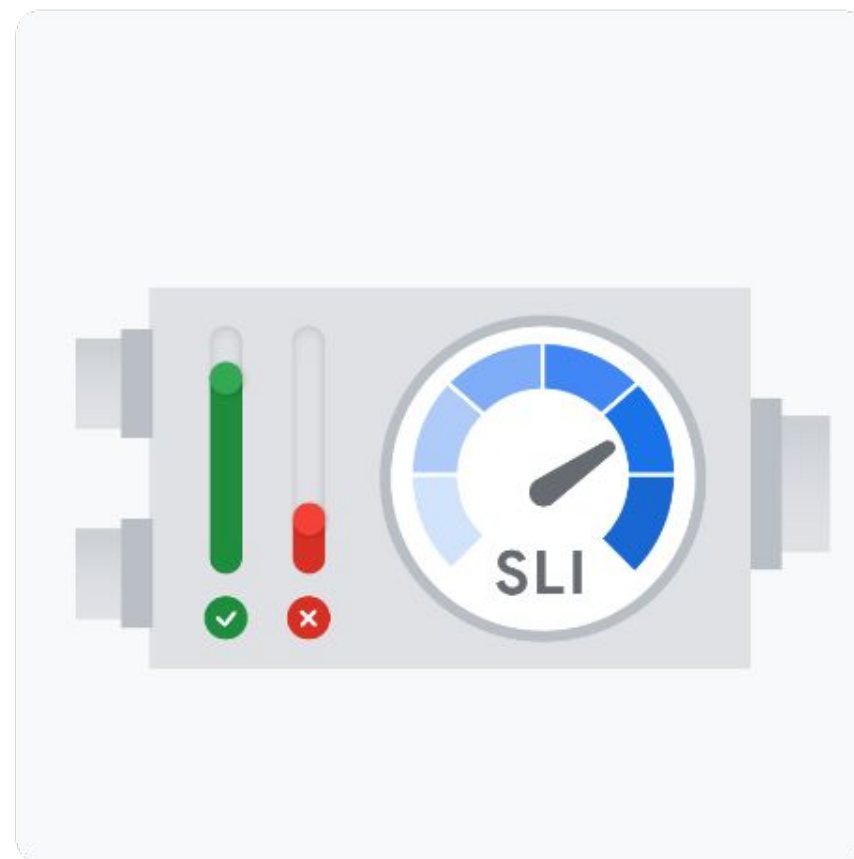
03

They can indicate service level objective violations.

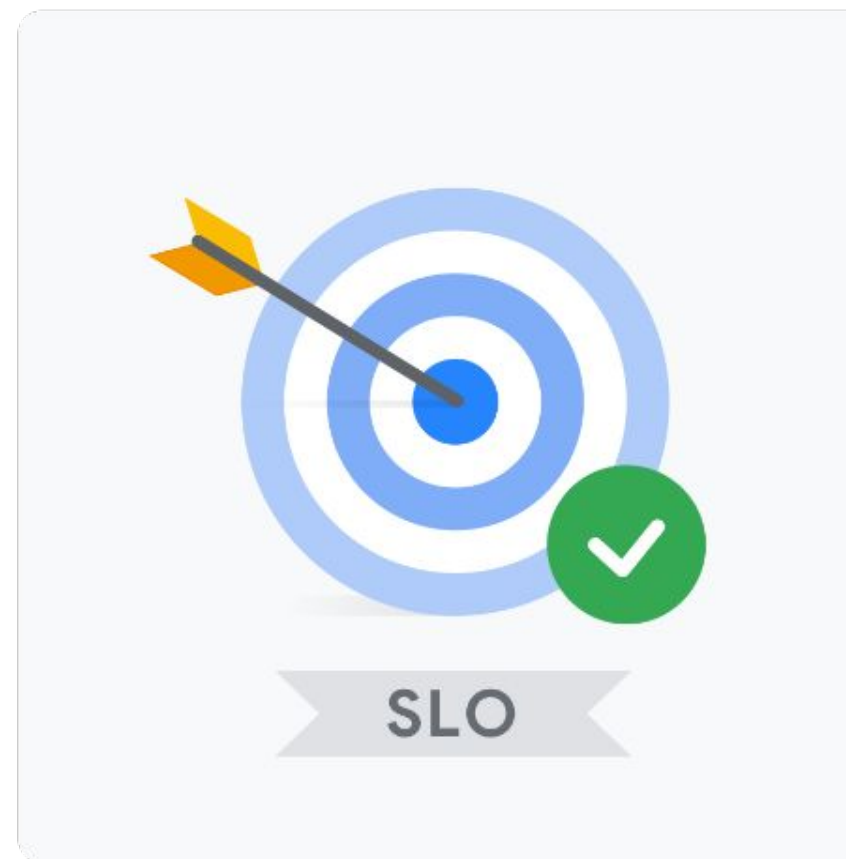
04

An error might mean it's time to send out an alert.

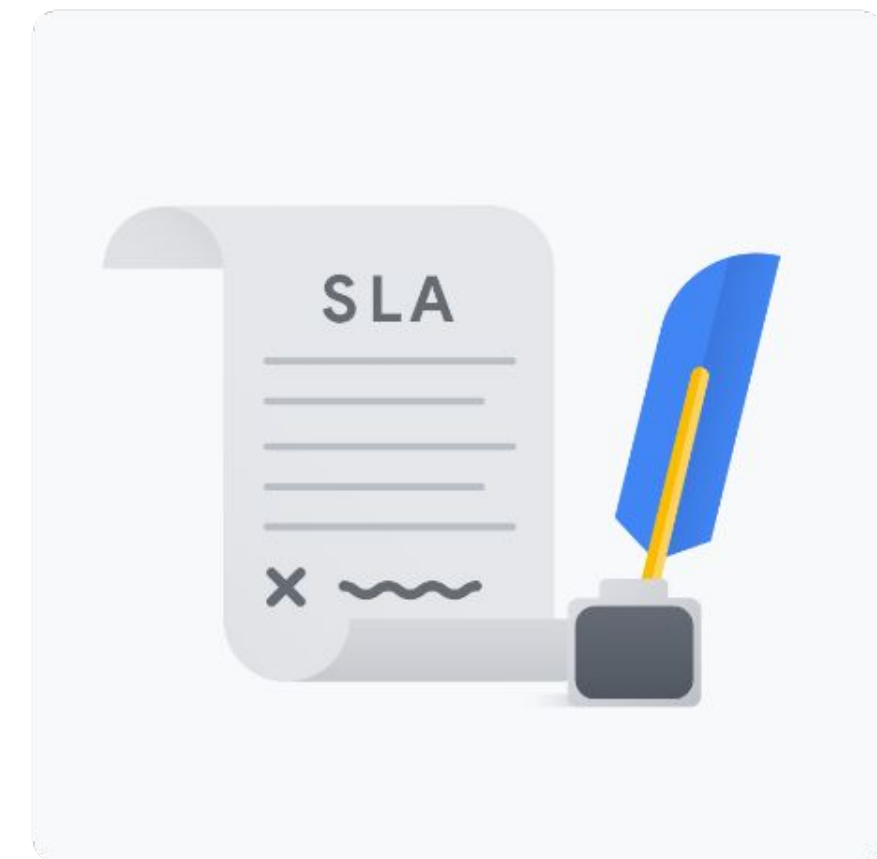
Three main concepts in site reliability engineering



Service level indicators
(SLIs)

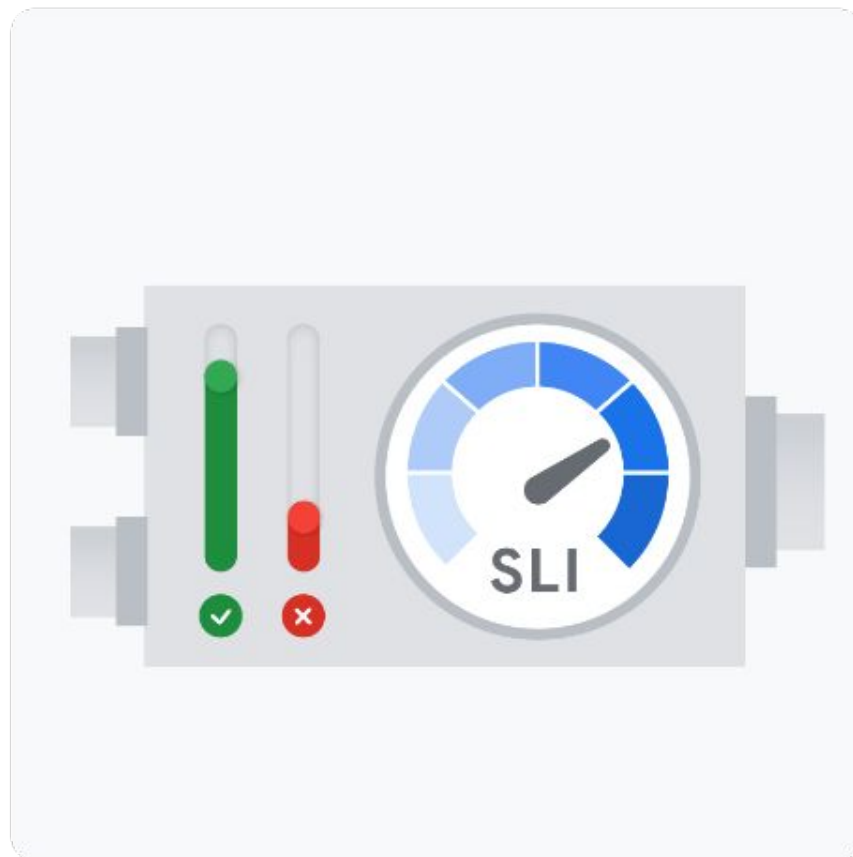


Service level objectives
(SLOs)



Service level agreements
(SLAs)

SLIs



Service-level indicators

Measurements that show how well a system or service is performing.

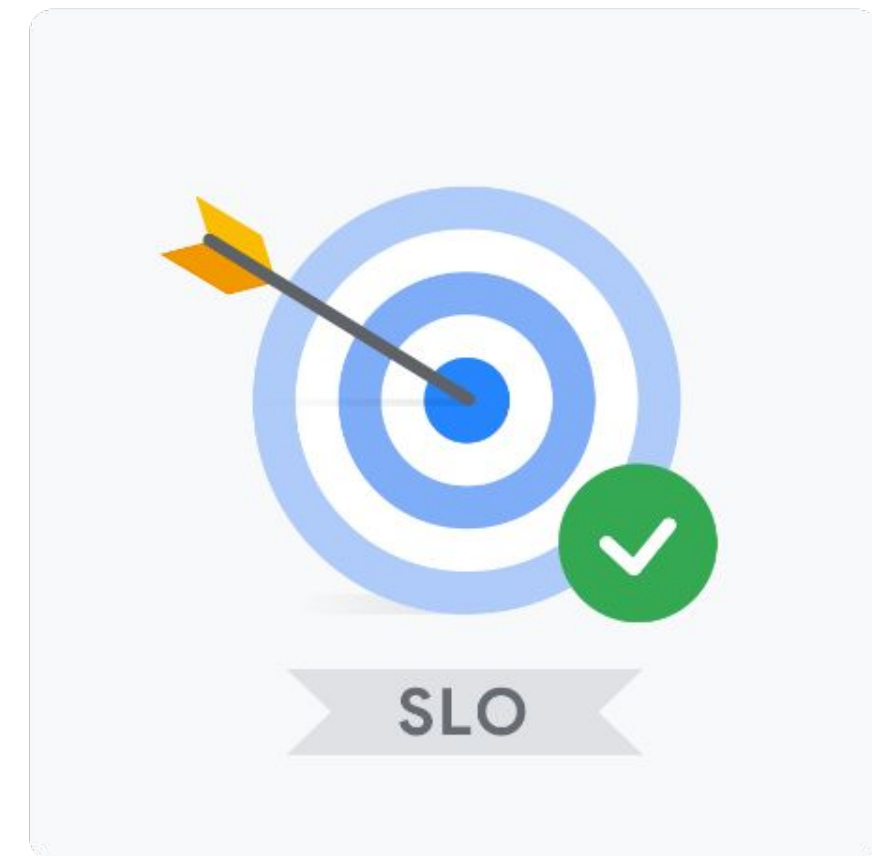
- They're specific metrics like:
 - Response time
 - Error rate
 - Percentage uptime

SLOs

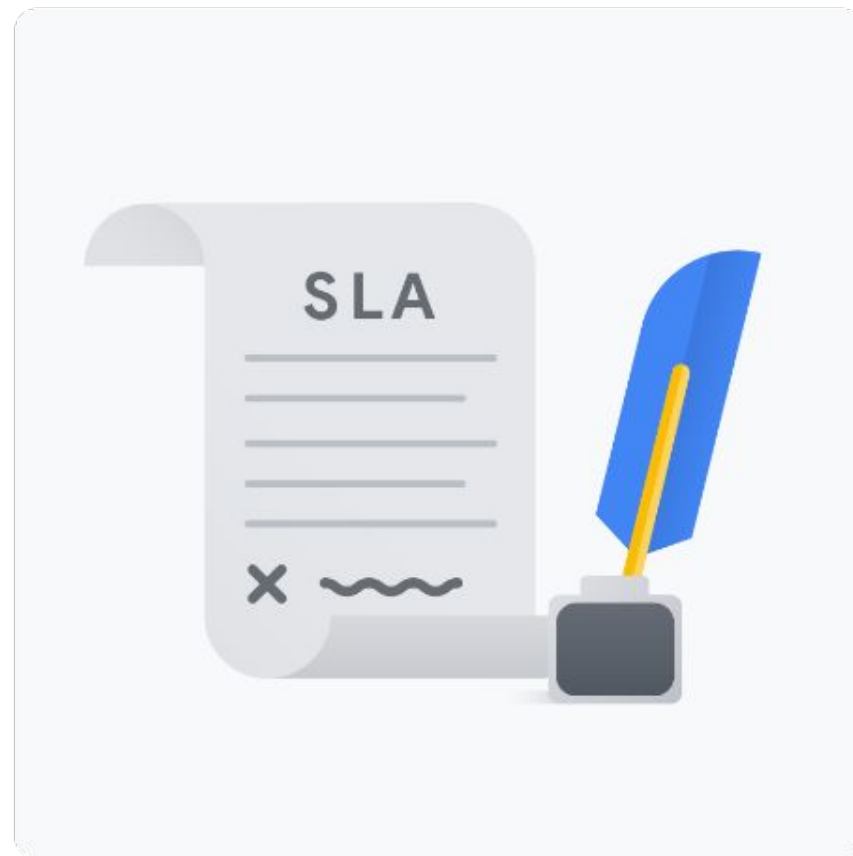
Service-level objectives

Goals that we set for a system's performance based on SLIs.

- They define what level of reliability or performance that we want to achieve.
 - For example, an SLO might state that the system should be available for 99.9% of the time in a month.



SLAs



Service-level agreements

Agreements between a cloud service provider and its customers.

- They outline the promises and guarantees regarding the quality of service.
- They include the agreed-upon SLOs, performance metrics, uptime guarantees, and any penalties or remedies if the provider fails to meet those commitments.

Quiz

Question

One of the four golden signals is **latency**. What does latency measure?

- A. How many requests reach a system
- B. How long it takes for a particular part of a system to return a result
- C. How close to capacity a system is
- D. System failures or other issues

Quiz

Answer

One of the four golden signals is **latency**. What does latency measure?

- A. How many requests reach a system
- B. How long it takes for a particular part of a system to return a result
- C. How close to capacity a system is
- D. System failures or other issues



Quiz

Question

Which metric shows how well a system or service is performing?

- A. Service-level indicators
- B. Service-level agreements
- C. Service-level contracts
- D. Service-level objectives

Quiz

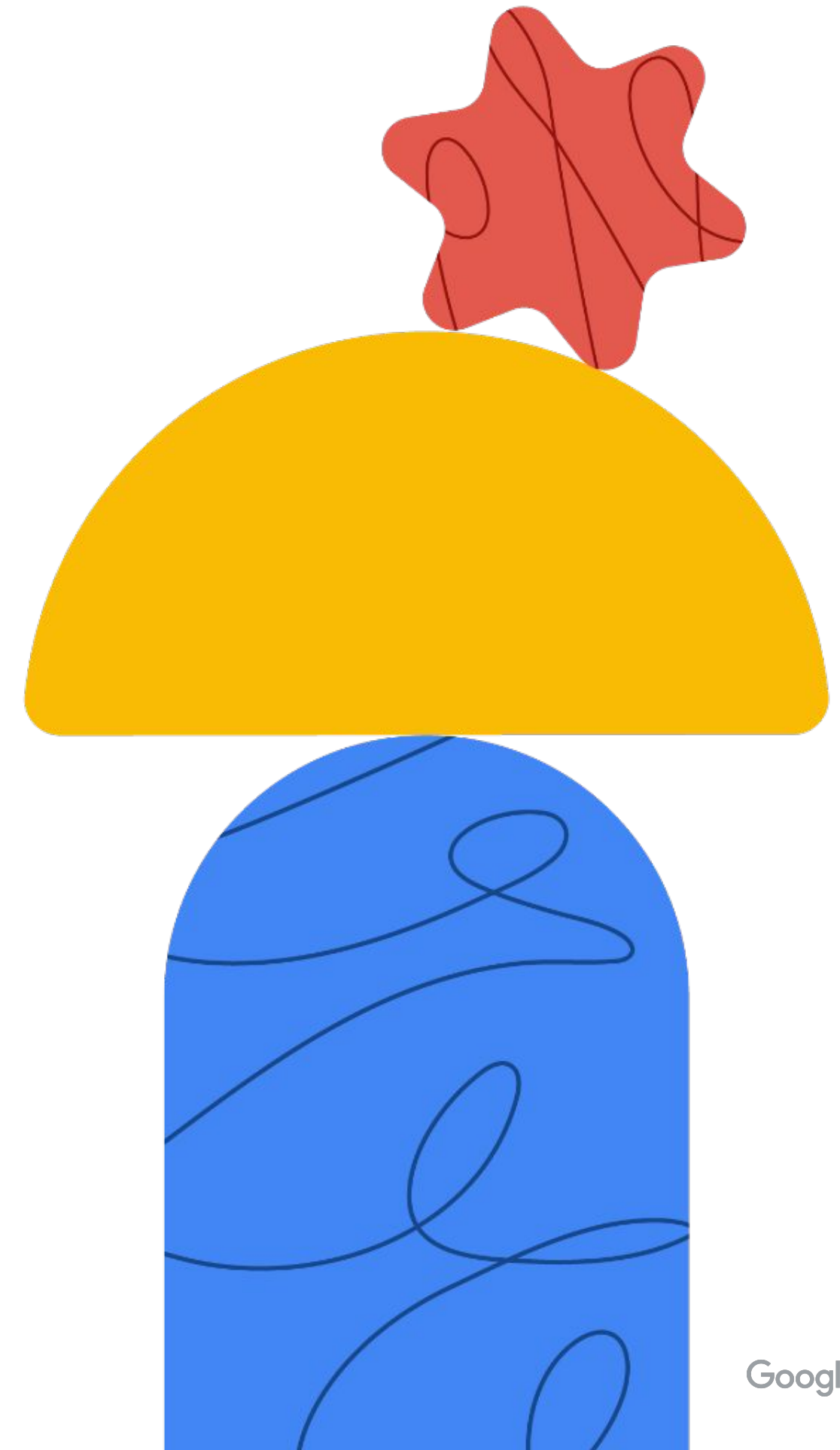
Answer

Which metric shows how well a system or service is performing?

- A. Service-level indicators
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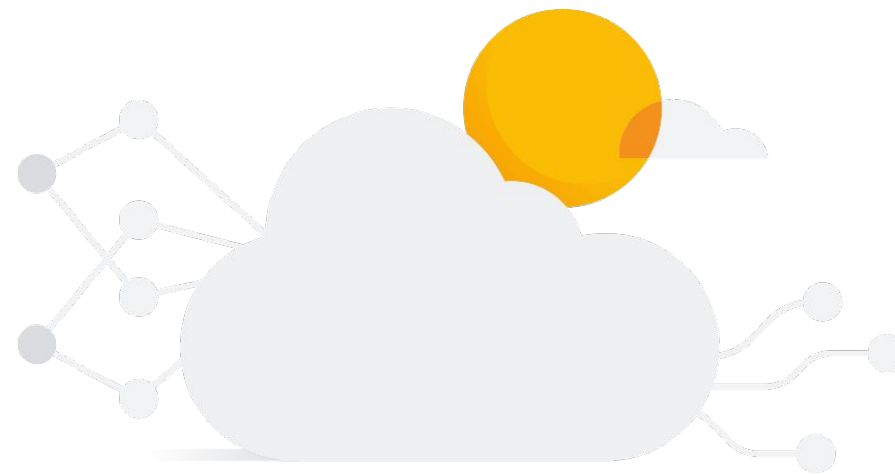
Designing resilient infrastructure and processes



Cloud infrastructure and processes must be resilient, fault-tolerant, and scalable

High availability

The ability of a system to remain operational and accessible for users even if hardware or software failures occur

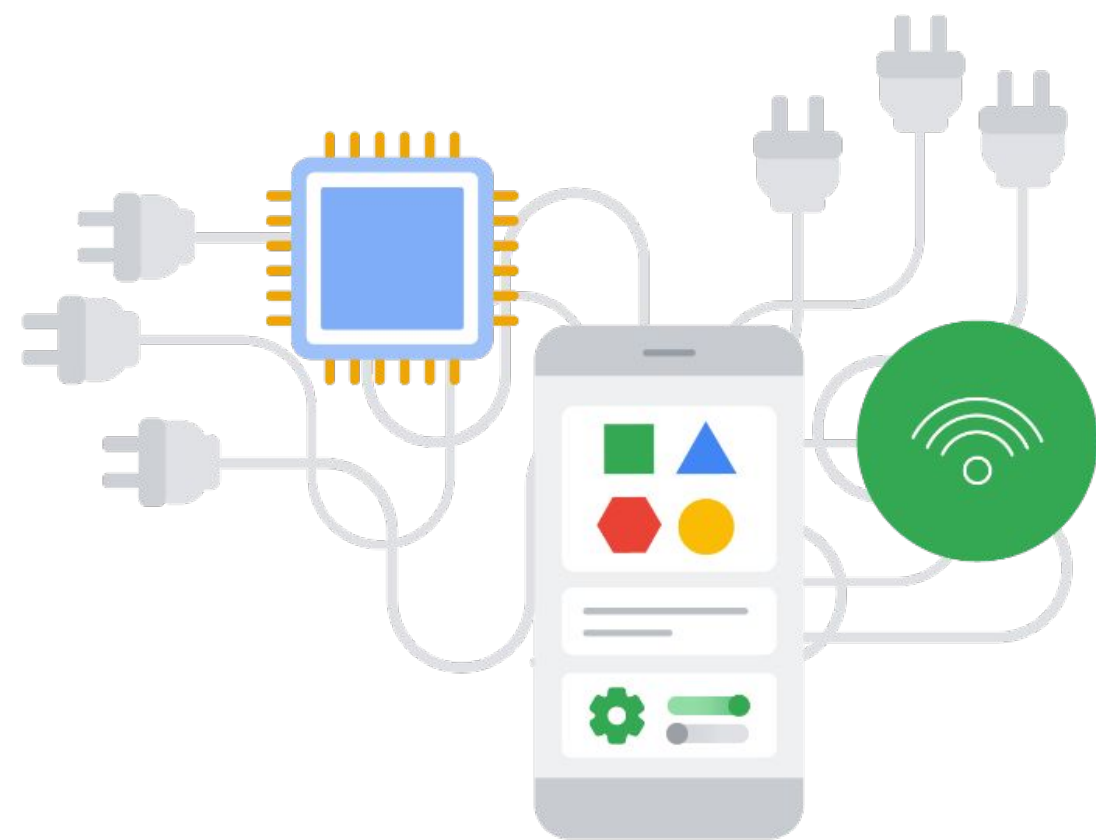


Disaster recovery

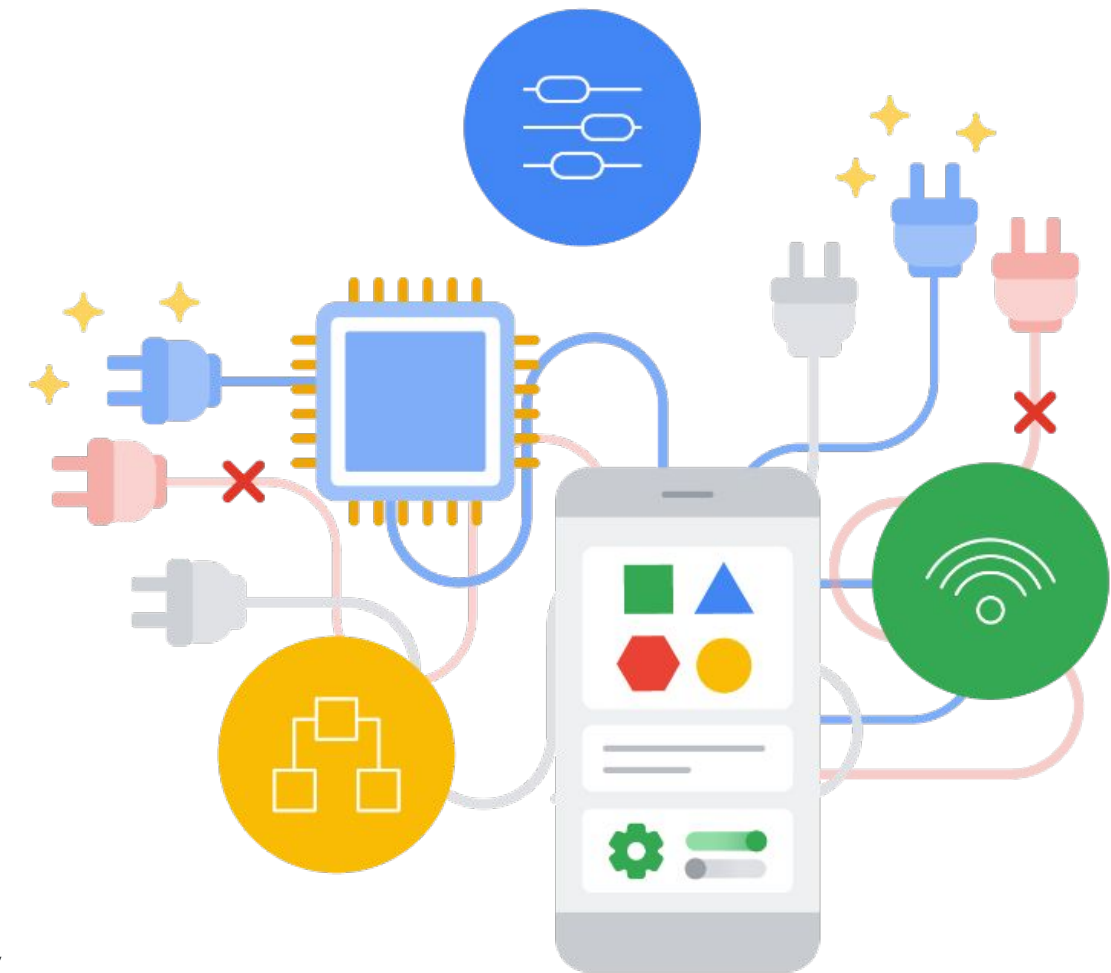
The process of restoring a system to a functional state after a major disruption or disaster

Redundancy

Duplicating critical components or resources to provide backup alternatives.



If one component fails,
a redundant component
can take over seamlessly.



Replication

Creating multiple copies of data or services and distributing them across different servers or locations.

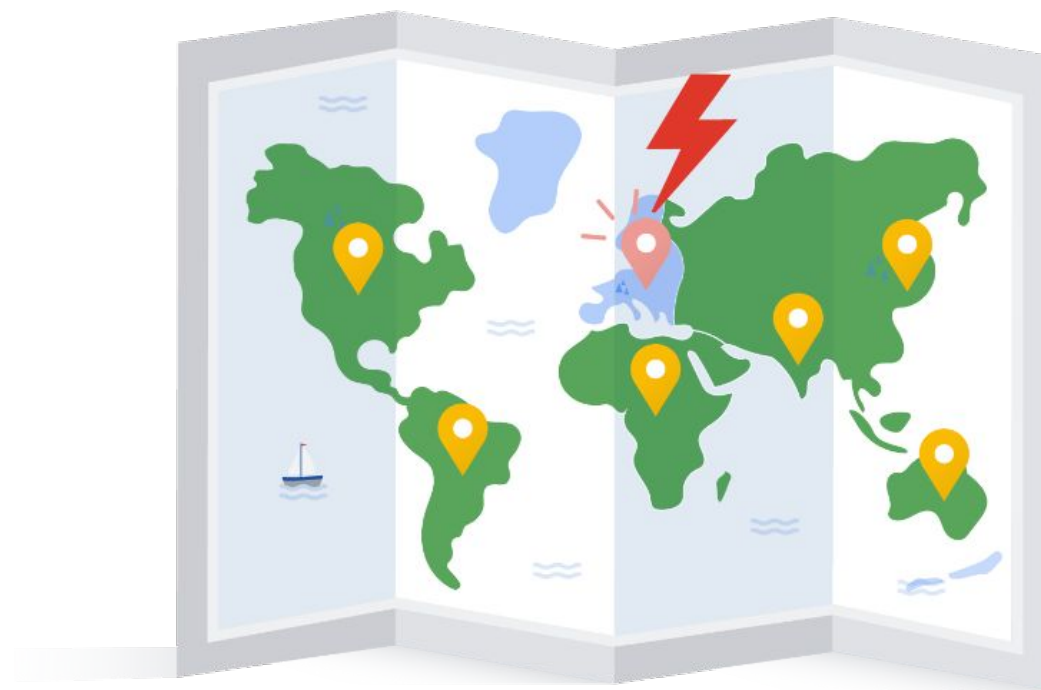


Replicating helps minimize the impact of hardware failures or outages and improves availability.



Regions

Cloud service providers offer multiple regions or data center locations spread across different geographic areas.

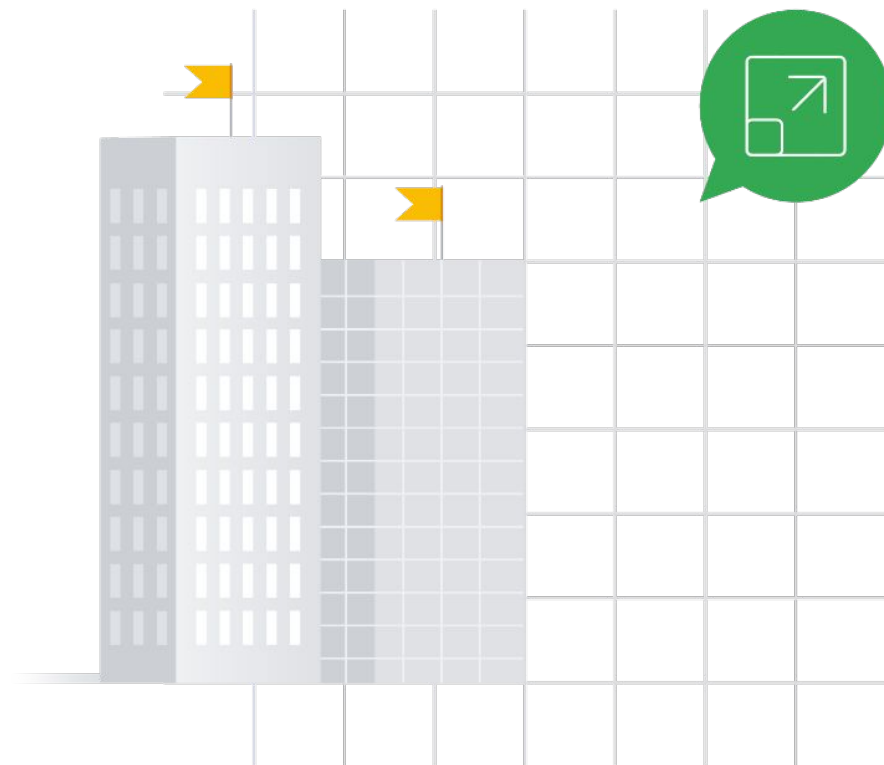


Ensures that if an entire region becomes unavailable, services can continue running from another region.

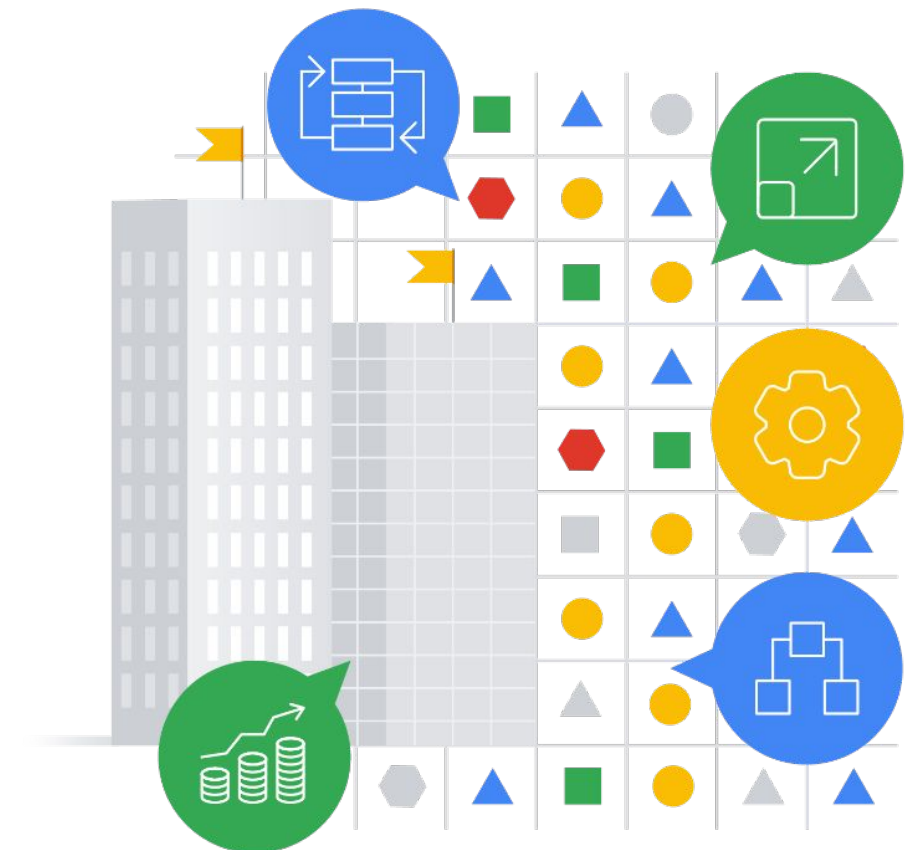


Scalable infrastructure

Allows organizations to handle varying workloads and accommodate increased demand without compromising performance or availability.

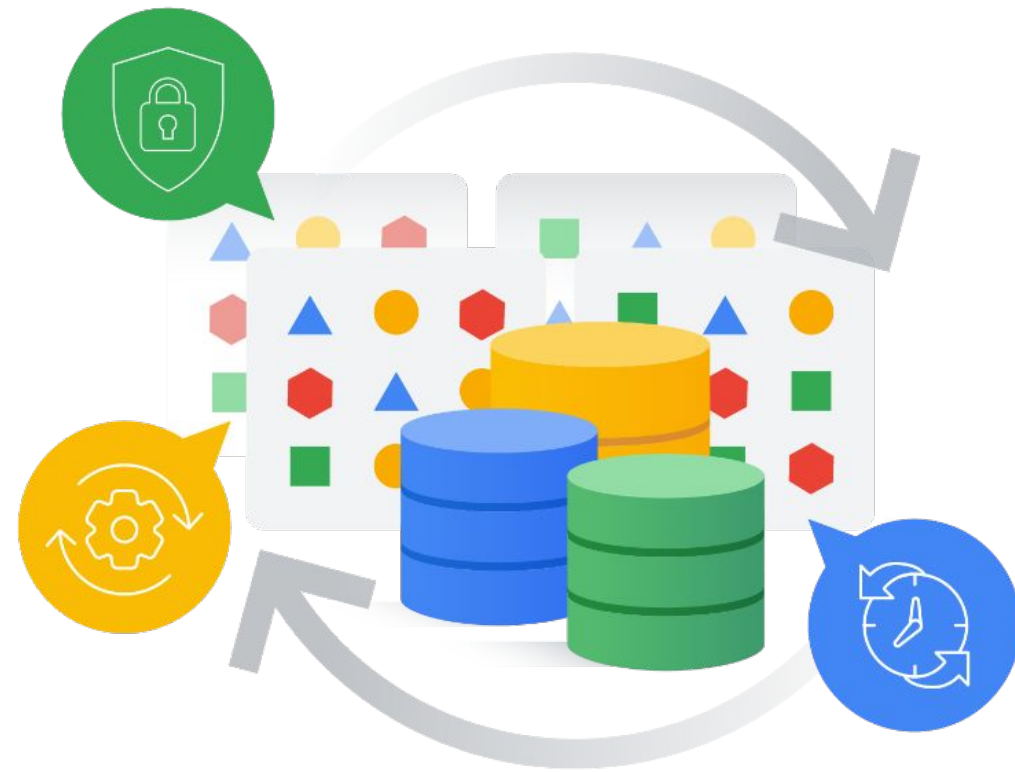


Cloud technologies enable the dynamic allocation and deallocation of resources.



Backups

Regular backups ensure that if data loss, hardware failures, or cyber-attacks occur, organizations can restore their systems to a previous state.

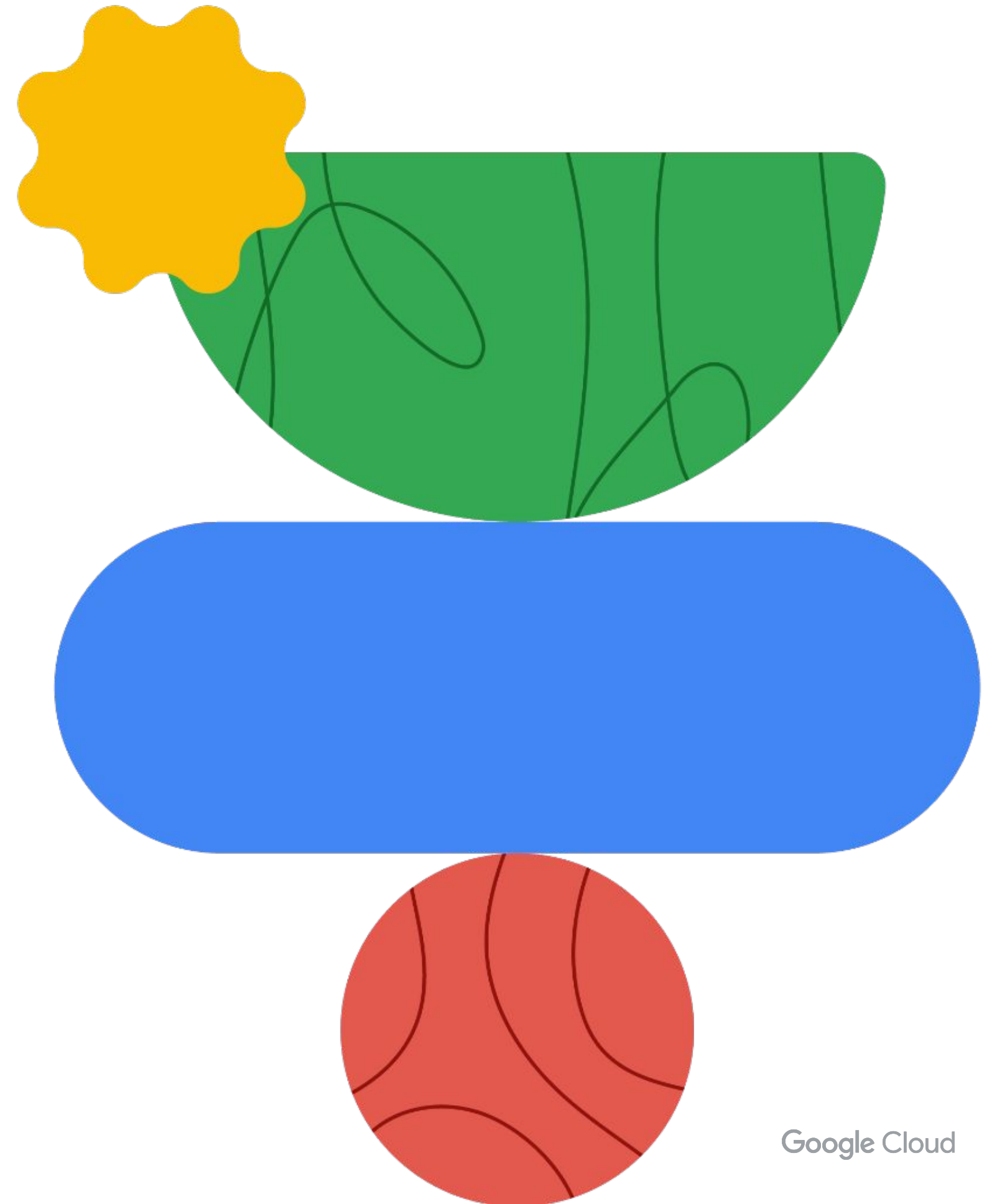


Cloud providers often offer backup services.



Backups should be stored in geographically separate locations.

Modernizing operations using Google Cloud



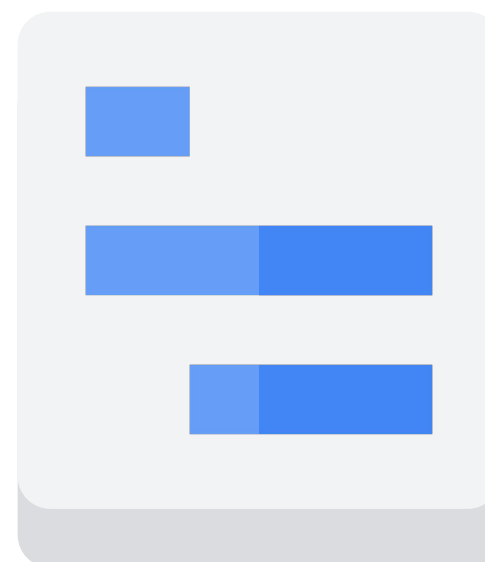
Google Cloud's operations suite



Cloud
Monitoring



Cloud
Logging



Cloud
Trace



Cloud
Profiler



Error
Reporting

Cloud Monitoring



Cloud
Monitoring

01

Provides a comprehensive view of cloud infrastructure and applications.

02

Collects metrics, logs, and traces from applications and infrastructure, and provides insights into their performance, health, and availability.

03

Lets you create alerting policies to notify when metrics, health check results, and uptime check results meet specified criteria.

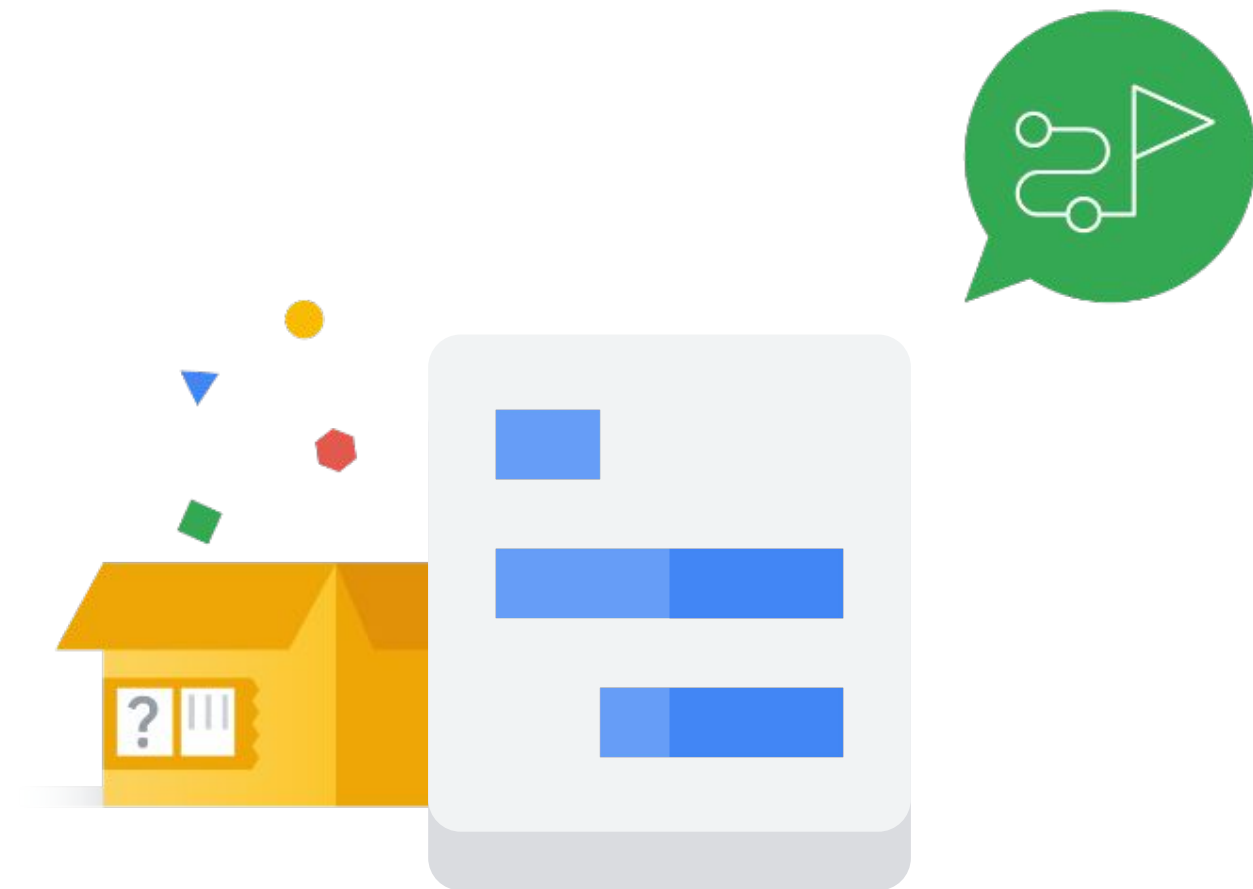
Cloud Logging



Cloud
Logging

- 01 Collects and stores all application and infrastructure logs.
- 02 Real-time insights to help you troubleshoot issues, identify trends, and comply with regulations.

Cloud Trace



Cloud
Trace

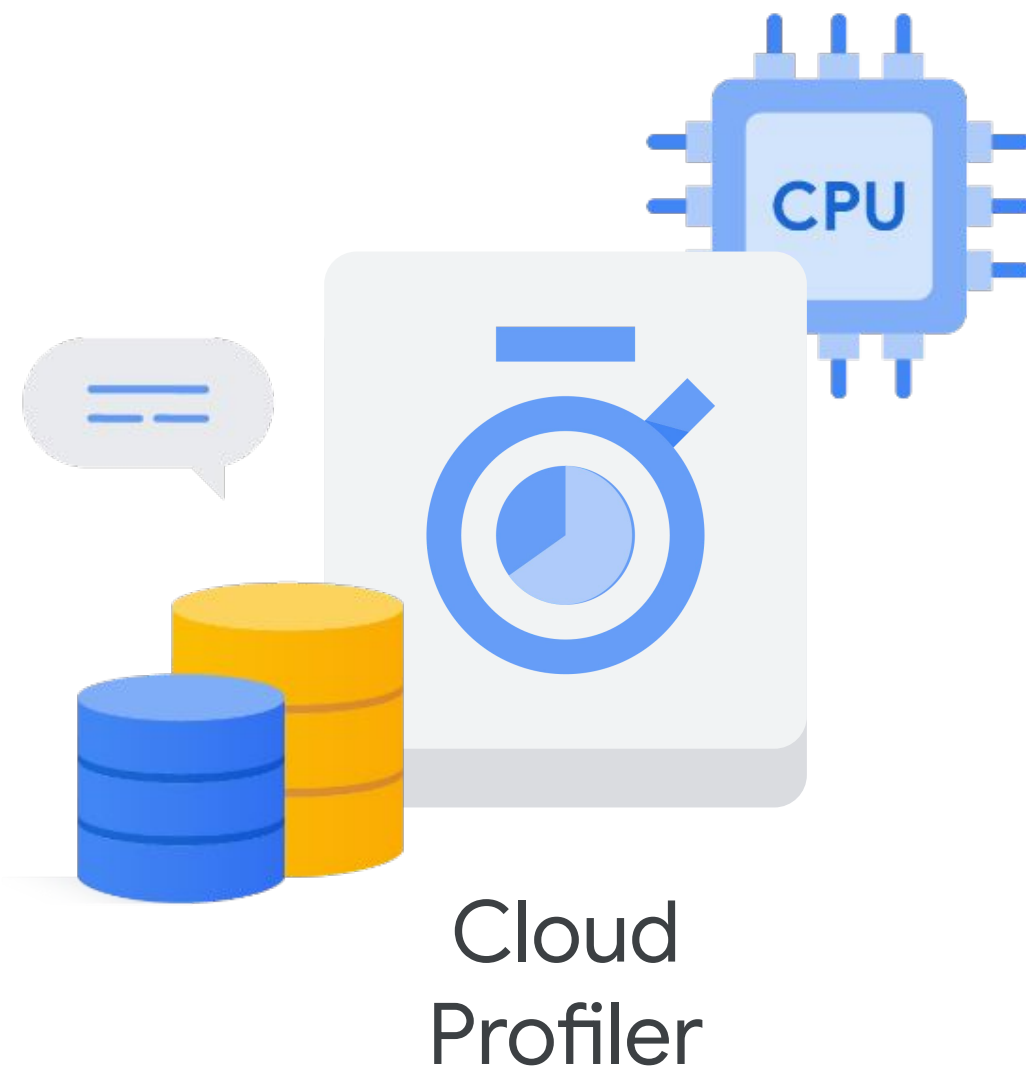
01

Helps identify performance bottlenecks in applications.

02

Collects latency data from applications, and provides insights into how they're performing.

Cloud Profiler



01

Identifies how much CPU power, memory, and other resources an application uses.

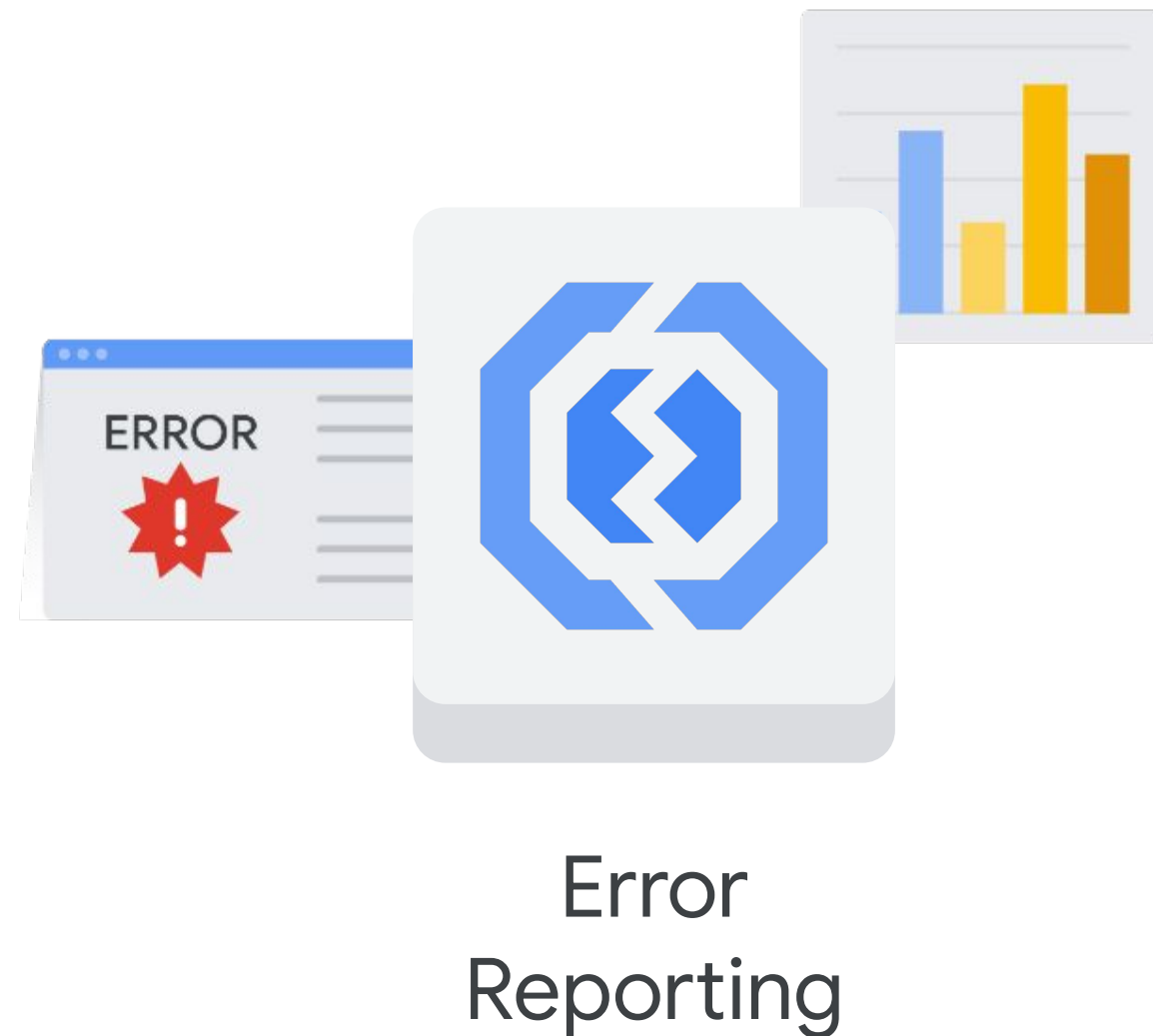
02

Continuously gathers CPU usage and memory-allocation information from production applications.

03

Provides insights into how applications are using resources.

Error Reporting



01

Error Reporting counts, analyzes, and aggregates the crashes in running cloud services in real-time.

02

A centralized error management interface displays the results with sorting and filtering capabilities.

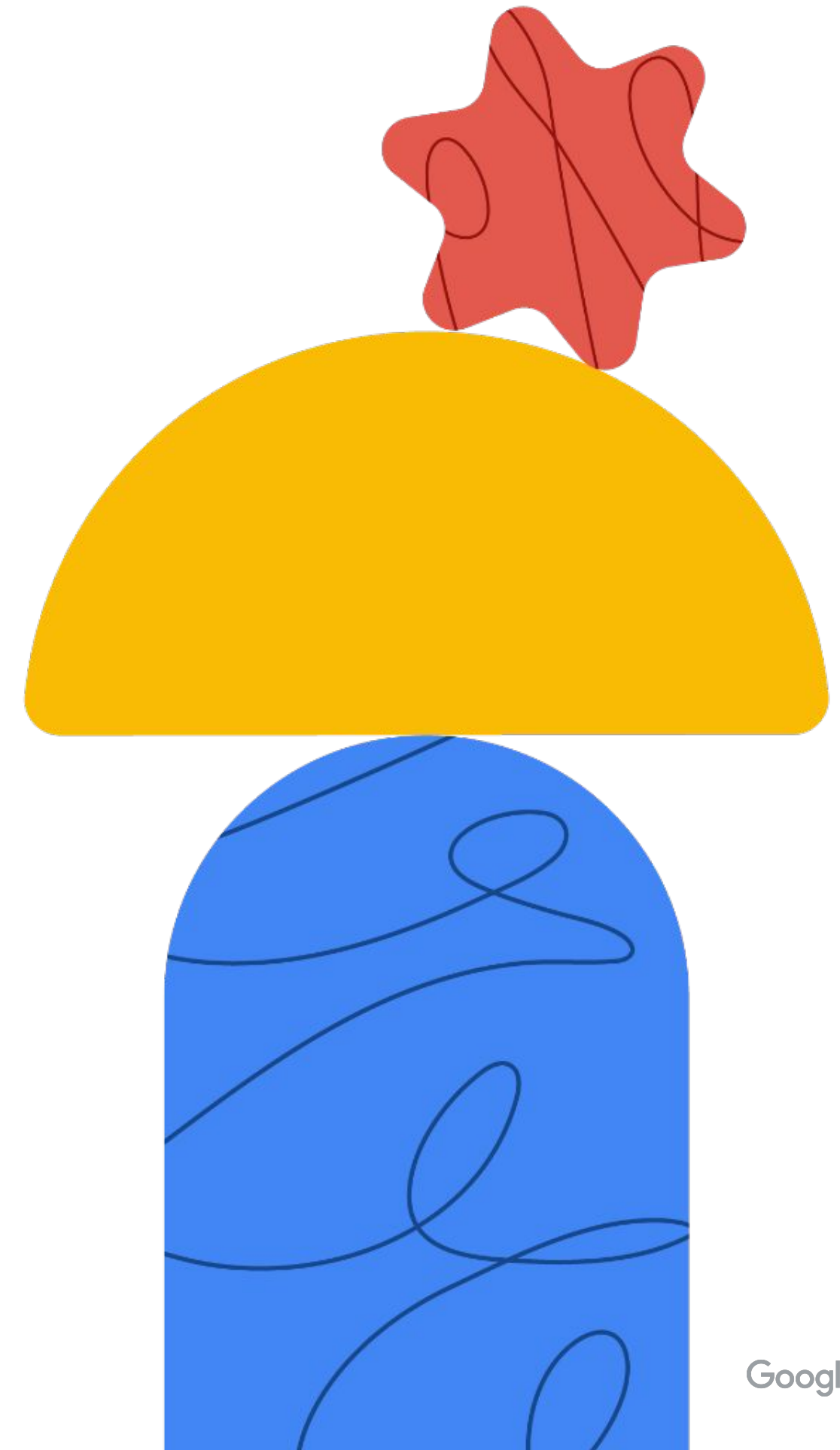
03

A dedicated view shows the error details: time chart, occurrences, affected user count, first- and last-seen dates.

04

Error Reporting supports email and mobile notifications through its API.

Google Cloud Customer Care



The four levels of Google Cloud Customer Care



Basic

Standard

Enhanced

Premium

Basic Support

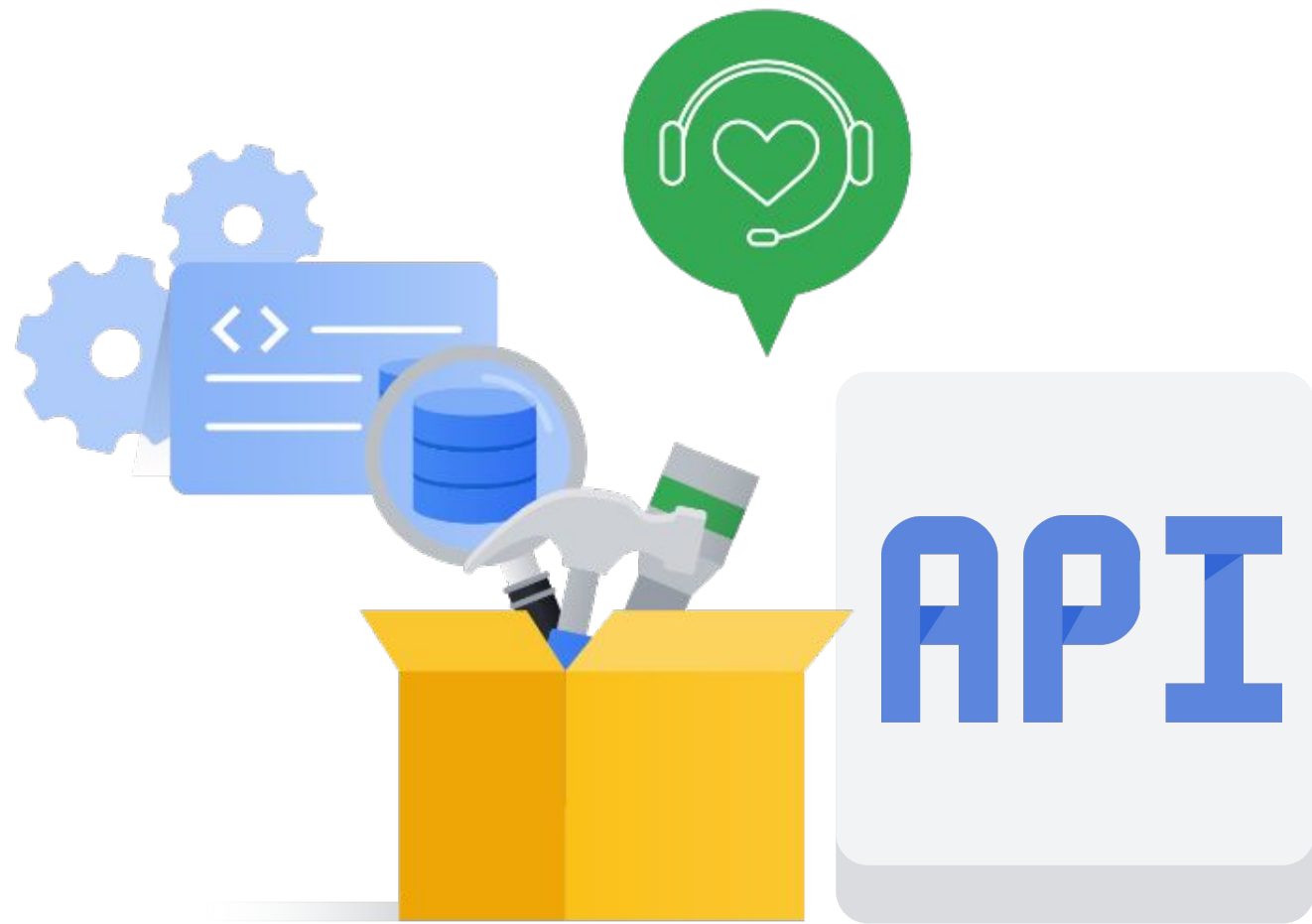


Is free and included for all Google Cloud customers.



Provides access to documentation, community support, Cloud Billing Support, and Active Assist recommendations.

Standard Support



Cloud
Support API



Recommended for workloads under development.



Provides unlimited access to tech support, which lets you troubleshoot, test, and explore.

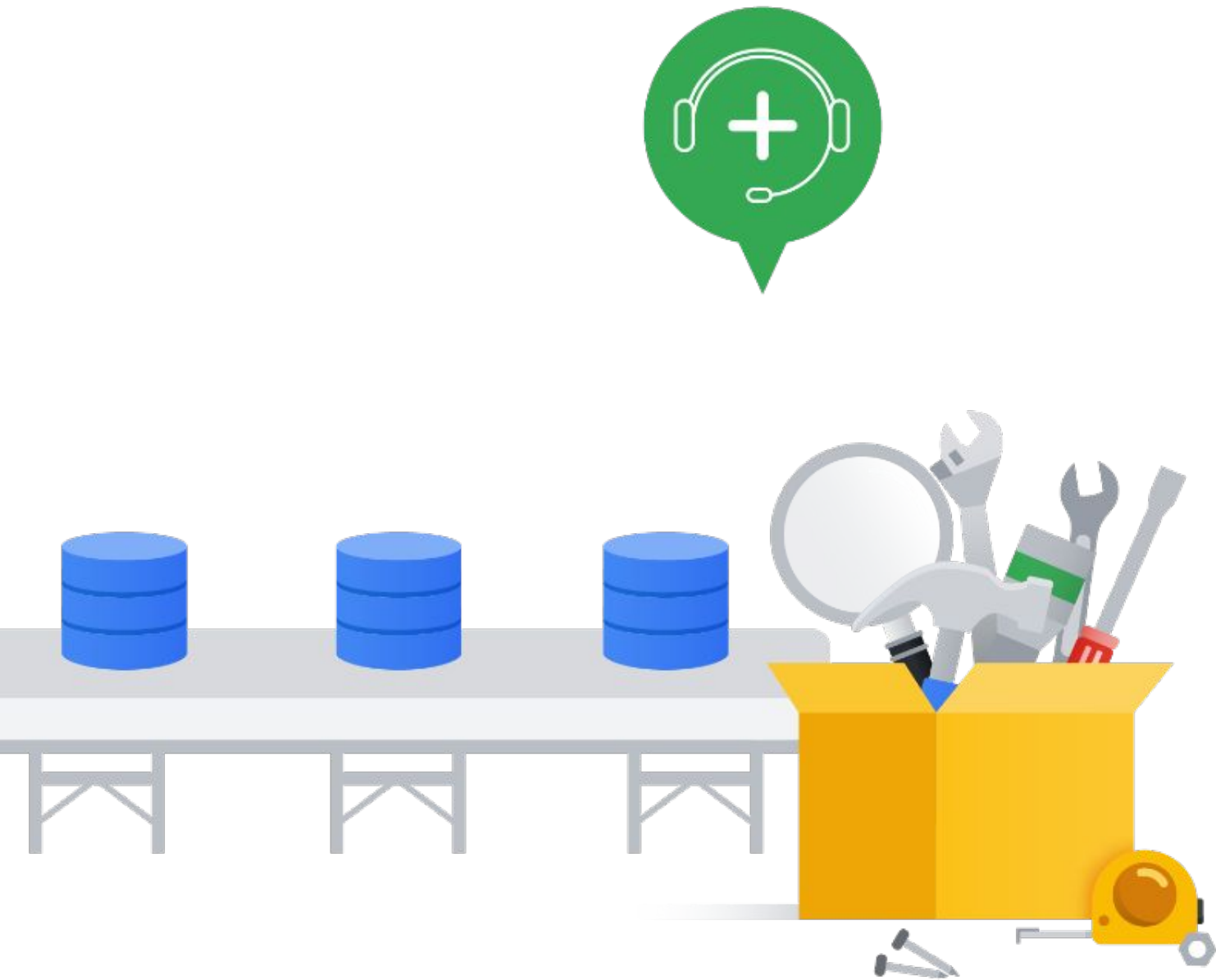


Offers unlimited individual access to English-speaking support representatives during working hours, five days a week.



Provides access to the Cloud Support API.

Enhanced Support



Designed for workloads in production, with fast response times and additional services.

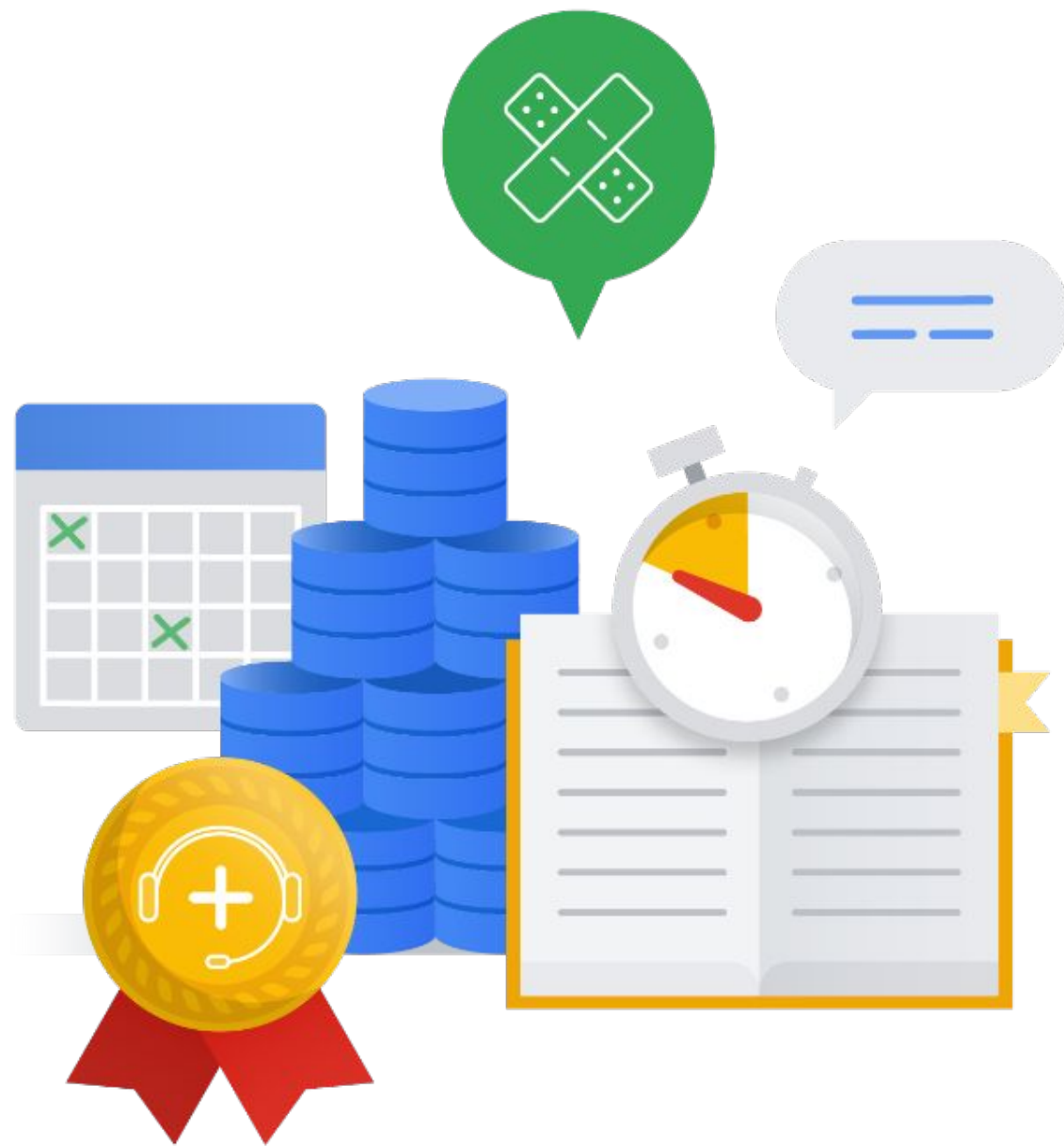


Available 24/7 in a selection of languages, and initial response times are quicker than those provided by Standard Support.



Offers technical support escalations and third-party technology support to help resolve multi-vendor issues.

Premium Support

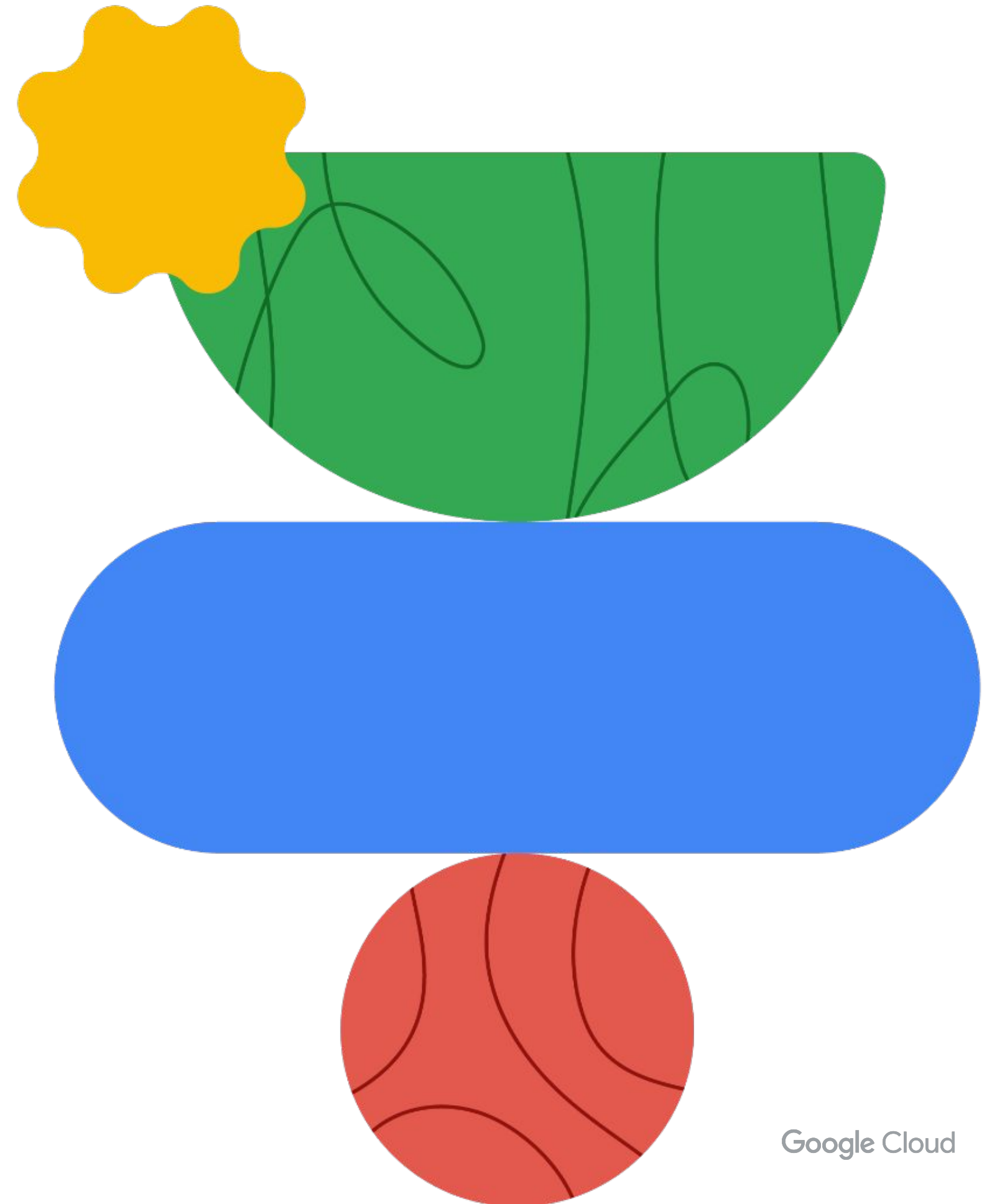


- ✓ Designed for enterprises with critical workloads.
- ✓ Features the fastest response time, Customer Aware Support, and a dedicated Technical Account Manager.
- ✓ Includes credit for the Google Cloud Skills Boost training platform.
- ✓ Includes Event Management Service for planned peak events.
- ✓ Includes Operational Health Reviews.
- ✓ Includes Customer Aware Support.

Google Cloud Customer Care

cloud.google.com/support

The life of a support case



The life of a support case



Quiz

Question

Google Cloud's operations suite provides a comprehensive set of monitoring, logging, and diagnostics tools. Which tool collects latency data from applications and provides insights into how they're performing?

- A. Cloud Profiler
- B. Cloud Monitoring
- C. Cloud Trace
- D. Cloud Logging

Quiz

Answer

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Quiz

Question

Which Google Cloud Customer Care support level is designed for enterprises with critical workloads and features the fastest response time?

- A. Standard Support
- B. Premium Support
- C. Basic Support
- D. Enhanced Support

Quiz

Answer

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Module 6

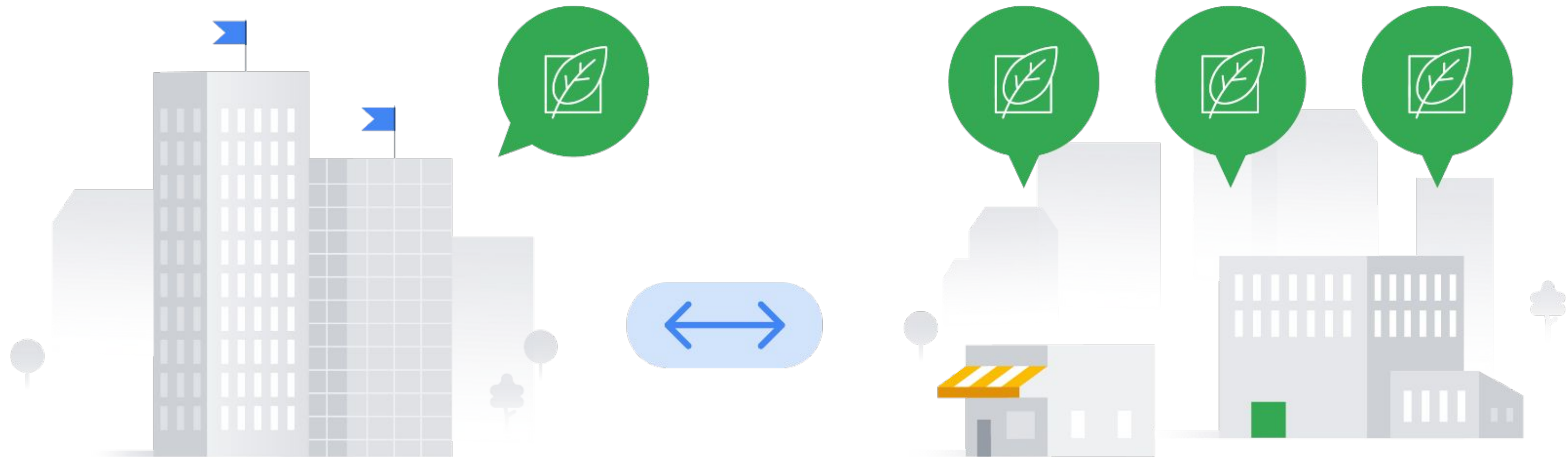
Scaling with Google Cloud Operations



Lessons

- 01 Financial governance and managing cloud costs
- 02 Operational excellence and reliability at scale
- 03 Sustainability with Google Cloud

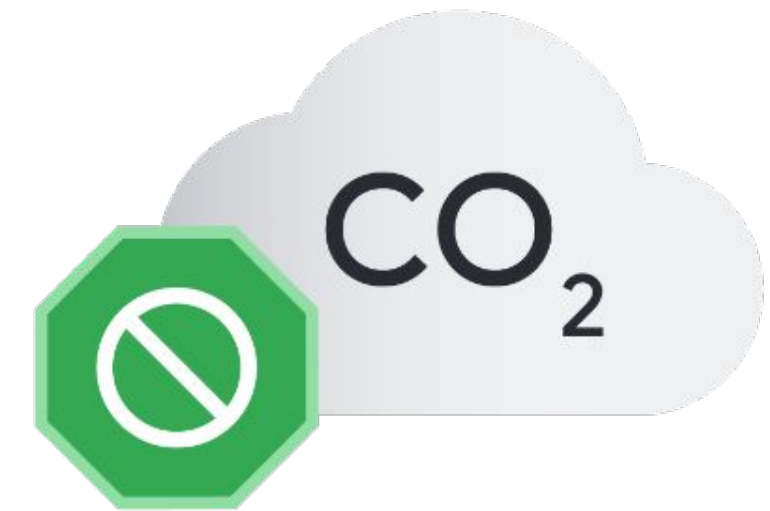
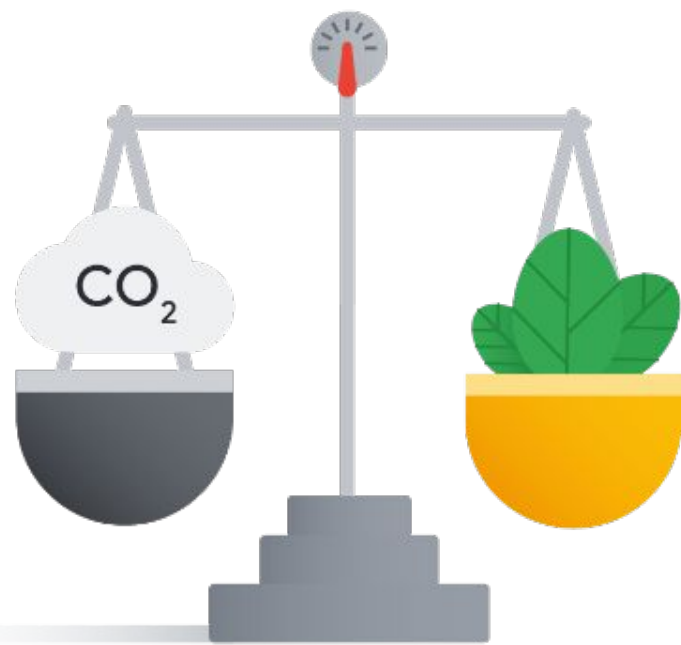
Just like our customers, Google tries to take care of the planet



Google's data centers were
the first to achieve
ISO 14001 certification.

Our customers

We have focused on the environment from the beginning



Founding decade
Carbon neutral

Second decade
Renewable energy

2030
Carbon-free

Quiz

Question

What sustainability goal does Google aim to achieve by the year 2030?

- A. To be the first major company to achieve 100% renewable energy.
- B. To be the first major company to be carbon neutral.
- C. To be the first major company to run its own wind farm.
- D. To be the first major company to operate completely carbon free.

Quiz

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Quiz

Question

Google's data centers were the first to achieve ISO 14001 certification. What is this standard's purpose?

- A. It's a framework for an organization to enhance its environmental performance through improving resource efficiency and reducing waste.
- B. It's a framework for identifying, predicting, and evaluating the environmental impacts of a proposed project.
- C. It's a framework for carbon footprinting that calculates the total amount of greenhouse gas emissions associated with a product, service, or organization.
- D. It's a framework for sustainable procurement, which is the process of purchasing goods and services in a way that minimizes environmental and social impacts.

Quiz

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Google Cloud